The Geometry of Uncertainty: Geodesic Bayesian Smoothing via Metric–Drift Disentanglement

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Abstract

Latent SDE smoothers and continuous-time inference pipelines often tune both drift and diffusion scale, blurring geometry with dynamics and complicating uncertainty quantification. We show that diffusion alone fixes the Onsager–Fokker Riemannian metric (transport cost) of Bayesian smoothing, while at fixed diffusion the drift is identifiable only up to a ρ-divergence-free gauge; the Schrödinger–KL principle selects a unique minimal-energy representative. Treating the diffusion scale σ as an inverse temperature $\beta{=}\sigma^{-2}$, we characterise the posterior's transition from deterministic prior flows ($\sigma{\to}0$) to time-decoupled evidence tilts ($\sigma{\to}\infty$, under mixing), with a mutual-information order parameter quantifying cross-time coupling. Recipe: set diffusion first (geometry), then select the canonical drift by minimal control energy. We focus on smoothing or trajectory coherence rather than marginal fitting under fixed diffusion schedules.

1. Introduction

In the last decade, the intersection of stochastic analysis and machine learning has witnessed a renaissance, driven by the parametrisation of latent dynamics via Stochastic Differential Equations (SDEs). This paradigm has revolutionised generative modelling through score-based diffusion models (Sohl-Dickstein et al., 2015; Song & Ermon, 2019; Song et al., 2021; Ho et al., 2020) (typically with a fixed forward diffusion schedule), enabled continuous-depth inference via Neural ODEs or SDEs (Chen et al., 2018; Rubanova et al., 2019; Li et al., 2020), and revived the classical Schrödinger Bridge problem for data interpolation (Léonard, 2014; Mikami & Thieullen, 2006; De Bortoli et al., 2021; Liu et al., 2022; Vargas et al., 2021; Carlier et al., 2023). By scaling to high dimensions, these methods have become the standard for modelling complex time-series and transport phenomena, effectively bridging the gap between data-driven learning and rigorous stochastic control. In linear-Gaussian state-space models, classical Kalman filtering and Rauch–Tung–Striebel (RTS) smoothing (Kalman, 1960; Rauch et al., 1965; Bell, 1994; Särkkä, 2008; 2013) provide closed-form posteriors that naturally respect the geometric separation between process noise and dynamics. Yet as architectures move to nonlinear and neural-parametrised regimes, this structural clarity is often lost. Two modelling layers are commonly blurred: *geometry*—diffusion $D(x,t)$ (or a diffusion schedule) fixes the Onsager–Fokker metric, i.e. the local transport cost; and *dynamics*—at fixed diffusion, the drift only specifies an admissible control and tangent through Fokker–Planck dynamics and is non-identifiable from time-marginals up to ρ–divergence-free gauge directions (Nelson, 1967; Jordan et al., 1998; Benamou & Brenier, 2000; Otto, 2001; Ambrosio et al., 2008; Villani, 2009). Conflation means treating changes in D as "dynamics" changes, and treating learned drifts as uniquely interpretable from marginal objectives. Our conclusions therefore concern (i) drift identifiability under the *same* diffusion geometry, and (ii) the consequences of changing diffusion (a geometry change). We primarily target smoothing, latent-SDE, and variational-inference settings (Blei et al., 2017; Naesseth et al., 2018) that treat diffusion as a learnable degree of freedom for uncertainty; for score-based models with fixed diffusion schedules, our message concerns geometric coherence of trajectories and bridges, not marginal matching.

To resolve this ambiguity, we propose **Geodesic Bayesian Smoothing**, characterising the exact posterior as a *Schrödinger–Doob transform* (Föllmer, 1988; Léonard, 2014) with time-sliced likelihood potentials (Proposition 1). This representation unlocks two geometric contributions and theory-driven diagnostics (for related work, see Appendix A):

1. **Drift as a gauge (main theorem).** At fixed diffusion geometry, drift is identifiable from marginals only up to ρ–divergence-free directions; the Schrödinger–KL principle selects a unique canonical representative by minimal kinetic energy (Theorem 1).

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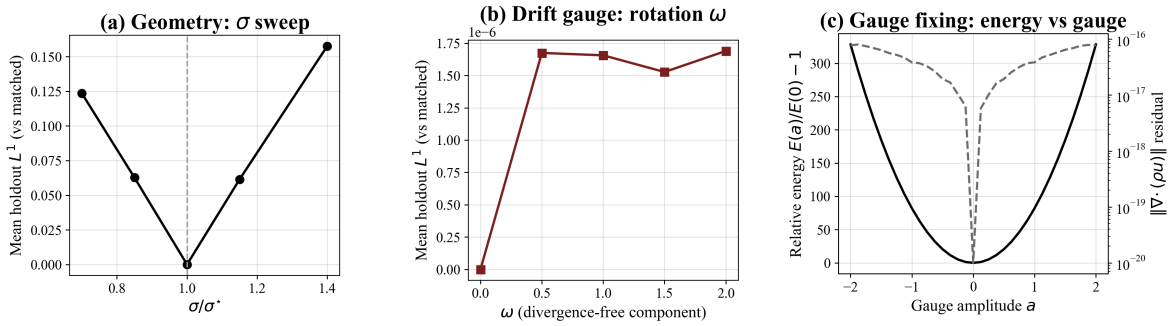

Figure 1. Teaser (a–c): diffusion sets smoothing geometry; at fixed diffusion drift is identifiable only up to a ρ-divergence-free gauge, and Schrödinger–KL selects the minimal-energy representative. Additional theorem-aligned diagnostics (RTS equivalence, σ temperature sweep, anchor-compatibility residual) are in Appendix C (Fig. 3).

2. **Geometry from diffusion, not drift (supporting results).** The kinetic term induces an Onsager–Fokker metric tensor depending only on the diffusion D (Proposition 2), and the posterior inherits a segment-wise geodesic structure via its endogenous anchors (Corollary 1).
3. **Corollaries and applications.** Treating isotropic diffusion σ as an inverse temperature, we quantify cross-time coupling by a mutual-information order parameter $\mathcal{I}_k(\sigma)$ (Corollary 4), and we validate the theorem-aligned gauge decomposition on a high-dimensional neural example.

Overall, the grid-exact smoother is meant as an *operational* modelling and training recipe and diagnostics toolbox: choose diffusion (geometry) first, then select the canonical drift within its gauge class, and validate with temperature sweeps, compatibility residuals, and gauge-controlled checks before scaling to neural SDE smoothers.

Roadmap. Sec. 2 fixes notation, the reference diffusion, and standard tools. Sec. 3 states the main results, centered on the gauge theorem. Sec. 4 collects corollaries and extensions (including diffusion-scale limits). Sec. 5 presents a high-dimensional neural application. Sec. 6 concludes; full proofs and implementation details are in the Appendix.

2. Preliminaries

2.1. Notation

Let $d \in \mathbb{N}$ and $T > 0$. Let $\Omega := C([0,T];\mathbb{R}^d)$ with coordinate process $X_t(\omega) = \omega(t)$. Write $\mathcal{P}(\Omega)$ for probability measures on Ω. For $Q \in \mathcal{P}(\Omega)$, write Q_t for its time-t marginal; when $Q_t \ll \mathrm{Leb}$ denote its density by ρ_t (and similarly $r(t,\cdot)$ for the reference marginal density).

We fix times $0 = t_0 < t_1 < \cdots < t_K \le T$ and abbreviate $\Delta t_k := t_k - t_{k-1}$. When we discuss controlled hard-anchor baselines we reserve ν_k for prescribed exogenous anchor marginals at the same time grid. For measures Q, P on a common measurable space, $D_{\mathrm{KL}}(Q\|P) := \int \log(\mathrm{d}Q/\mathrm{d}P)\,\mathrm{d}Q$ when $Q \ll P$, and $+\infty$ otherwise. For a symmetric positive-definite matrix M, we use $\langle a,b\rangle_M := a^\top M b$ and $\|a\|_M^2 := \langle a,a\rangle_M$. We write $\mathcal{P}_2(\mathbb{R}^d)$ for probability measures with finite second moment and W_2 for the quadratic Wasserstein distance.

2.2. Reference Dynamics

Our structural results extend to general dominated uniformly elliptic diffusions with strictly positive transition densities. As a concrete running reference we adopt an Ornstein–Uhlenbeck (OU) prior:

$$dX_t = -AX_t\,dt + \Sigma\,dW_t, \qquad X_0 \sim \rho_0, \tag{1}$$

and denote by $P_{\mathrm{ref}} \in \mathcal{P}(\Omega)$ its law on paths. We write $K_{s\to t}(x',x)$ for the (Lebesgue) transition density and $r(t,x)$ for the time-t marginal density, i.e. $P_{\mathrm{ref}}(X_t \in \mathrm{d}x) = r(t,x)\,\mathrm{d}x$. We assume non-degenerate diffusion: $D := \frac{1}{2}\Sigma\Sigma^\top \succ 0$, hence $K_{s\to t}$ and $r(t,\cdot)$ are strictly positive. On the fixed time grid, we abbreviate $K_k := K_{t_{k-1}\to t_k}$. When $\Sigma = \sigma I$ is isotropic, $D = (\sigma^2/2)I$ and we use $\sigma > 0$ as a diffusion-scale parameter.

2.3. Multi-time observations and per-time likelihood tilts

Assumption 1 (Multi-marginal observations)**.** *Likelihood factors $\ell(y_k \mid x) \ge 0$ at $\{t_1,\dots,t_K\} \subset (0,T]$ satisfy:*

(i) ***Factorisation:*** $L_y(X) := \prod_{k=1}^K \ell(y_k \mid X_{t_k}) = \exp(-\sum_{k=1}^K V_k(X_{t_k}))$, *with* $V_k := -\log \ell(y_k \mid \cdot) \in (-\infty,\infty]$.
(ii) ***Evidence finiteness:*** $0 < Z := \mathbb{E}_{P_{ref}}[L_y(X)] < \infty$.
(iii) ***Local normalisation:*** $0 < Z_{y_k} := \int r(t_k,x)\,\ell(y_k \mid x)\,\mathrm{d}x < \infty$ *for each k.*[1]

Posterior path measure. Define P_{post} as the Doob tilt of

[1] If $Z_{y_k} = 0$, the corresponding hard constraints are infeasible; overlap can be restored via support tempering.

P_{ref}:

$$\frac{\mathrm{d}P_{\text{post}}}{\mathrm{d}P_{\text{ref}}} = Z^{-1} L_y. \tag{2}$$

We denote its time marginals by $\mu_t := (P_{\text{post}})_t$ and write $\mu_k := \mu_{t_k}$.

Per-time likelihood tilts. Each single observation also induces a local Bayes update of the reference marginal:

$$\rho^{\text{obs}}_{t_k}(x) := \frac{r(t_k, x)\, \ell(y_k \mid x)}{Z_{y_k}}. \tag{3}$$

These per-time tilts are convenient *local* baselines (and can be used as *exogenous* anchors in bridge-style methods). For $K > 1$ they generally differ from the true posterior marginals μ_k due to cross-time coupling; see Proposition 3.

Standing assumptions and limitations. We assume a dominated, *uniformly elliptic* reference diffusion whose time–marginals and discretised transition kernels are strictly positive. This ensures the multi–time Schrödinger system is well posed and log-domain message passing is numerically stable; for hard-anchor baselines, strict positivity also yields linear IPFP convergence in Hilbert's projective metric (Appendix O). Learned references are compatible provided their diffusion is kept non-degenerate; hypoelliptic references require sub–Riemannian geometry (cf. Sec. 6).

2.4. Free-Energy Formulation for Posterior Path Measures

With P_{ref} and L_y fixed (Assumption 1), the posterior P_{post} is defined by the Gibbs tilt (2). To cast smoothing as an optimisation problem we consider candidate path measures $Q \ll P_{\text{ref}}$ and define the path-space free energy

$$\begin{aligned} \mathcal{F}[Q] &:= D_{\mathrm{KL}}(Q \| P_{\text{ref}}) - \mathbb{E}_Q[\log L_y(X)] \\ &= D_{\mathrm{KL}}(Q \| P_{\text{ref}}) + \sum_{k=1}^{K} \mathbb{E}_{Q_{t_k}}[V_k], \end{aligned} \tag{4}$$

where $V_k = -\log \ell(y_k \mid \cdot)$ and X denotes the canonical process under Q. Unless stated otherwise, our optimisation ranges over the *full* admissible class (no parametric restriction); approximations enter only after restricting Q.

Admissible drift perturbations (Girsanov). In diffusion settings we restrict to $Q \ll P_{\text{ref}}$ with the same diffusion coefficient. Under standard Girsanov conditions, such Q correspond to choosing an initial law $Q_0 \ll \rho_0$ and adding a progressively measurable control drift u_t to the reference drift. The path-space KL decomposes into an initial mismatch plus a quadratic kinetic energy:

$$\begin{aligned} D_{\mathrm{KL}}(Q \| P_{\text{ref}}) &= D_{\mathrm{KL}}(Q_0 \| \rho_0) \\ &\quad + \tfrac{1}{2}\, \mathbb{E}_Q \int_0^T u_t^\top (\Sigma\Sigma^\top)^{-1} u_t \, \mathrm{d}t. \end{aligned}$$

For isotropic $\Sigma = \sigma I$ this reduces to

$$D_{\mathrm{KL}}(Q \| P_{\text{ref}}) = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \frac{1}{2\sigma^2} \mathbb{E}_Q \int_0^T \|u_t\|^2 \, \mathrm{d}t.$$

Equivalently, writing $D := \frac{1}{2}\Sigma\Sigma^\top \succ 0$, the kinetic term is $\frac{1}{4}\, \mathbb{E}_Q \int_0^T \|u_t\|^2_{D^{-1}} \, \mathrm{d}t$. Substituting into (4) yields the free-energy/action form

$$\begin{aligned} \mathcal{F}[Q] = \underbrace{D_{\mathrm{KL}}(Q_0 \| \rho_0)}_{\text{Boundary Term}} &+ \underbrace{\frac{1}{4}\, \mathbb{E}_Q \int_0^T \|u_t\|^2_{D^{-1}} \, \mathrm{d}t}_{\text{Kinetic Energy}} \\ &+ \underbrace{\sum_{k=1}^{K} \mathbb{E}_{Q_{t_k}}[V_k]}_{\text{Potential Energy}}. \end{aligned} \tag{5}$$

This connects Bayesian smoothing to classical mechanics and optimal control: after fixing the boundary term, the kinetic part induces a Riemannian metric on probability measures (Otto geometry) once the drift gauge is fixed, which we make explicit in Sec. 3.1. When $Q_0 = \rho_0$ (as in our fixed-initial hard-anchor bridge baselines) the boundary term vanishes.

Standard tools. On discretised grids, posterior marginals are computed by the standard Doob–h / forward–backward recursion. For controlled baselines with prescribed anchors, we solve the multi-marginal Schrödinger system via log-domain IPFP/Sinkhorn scaling. Throughout, message passing is performed in the log domain for numerical stability; details appear in Appendix N–O.

3. Main Results

We remain on the Bayesian posterior line: all constructions in this section start from the Gibbs tilt $P_{\text{post}} \propto L_y\, P_{\text{ref}}$ with fixed likelihood factors. When we introduce hard marginal constraints, it is only as a fixed-anchor baseline that exposes the kinetic geometry under clean fixed-endpoint constraints; the posterior inherits these statements by taking the anchors to be its own marginals.

3.1. Drift as a Gauge: Canonical Representative via Minimal Energy

At fixed diffusion geometry, marginal evolution does not identify drift uniquely: time-marginals only constrain a weighted divergence $\nabla \cdot (\rho\, \cdot)$. The Schrödinger–KL principle resolves this by selecting a canonical representative within each gauge class via minimal kinetic energy. Recall the action decomposition (5) from Sec. 2.4. Theorem 1 identifies the unique canonical drift selected by minimizing the kinetic term within each gauge class.

Drift gauge and canonical representative. Even with D fixed, the drift that realises a given marginal evolution

is not unique: any component invisible to the weighted divergence $\nabla\cdot(\rho\,\cdot)$ cannot be recovered from time-marginals. The correct notion of identifiability is therefore a gauge equivalence, and the Schrödinger–KL principle selects a unique *canonical* drift by minimising kinetic energy.

Definition 1 (Drift gauge equivalence at a fixed marginal)**.** Let ρ be a strictly positive probability density on $\mathbb{R}^d$ and let $D \succ 0$ be a symmetric diffusion matrix. Two vector fields $u, \tilde{u} \in L^2(\rho; D^{-1})$ are *gauge equivalent at* ρ, denoted $u \sim_\rho \tilde{u}$, if

$$\nabla\cdot(\rho u) = \nabla\cdot(\rho\tilde{u}) \quad \text{in the sense of distributions,}$$

equivalently $\nabla\cdot(\rho(u-\tilde{u})) = 0$. The induced tangent $\delta\rho := -\nabla\cdot(\rho u)$ depends only on the equivalence class $[u]_\rho$.

Theorem 1 (Gauge decomposition and canonical representative)**.** *Fix ρ and D as in Definition 1, and let $\delta\rho$ be any signed density with $\int \delta\rho\, dx = 0$. Assume either* (i) $\mathbb{R}^d$ *with sufficient decay so that integration by parts has no boundary terms, or* (ii) *a compact domain with reflecting or periodic boundary conditions. Define the affine constraint set and kinetic energy:*

$$\mathcal{U}_\rho(\delta\rho) := \left\{u \in L^2(\rho; D^{-1}) : \ \delta\rho = -\nabla\cdot(\rho u)\right\},$$
$$\mathcal{E}_\rho(u) := \tfrac{1}{4}\int \langle u, D^{-1}u\rangle\, \rho\, dx.$$

Then the following hold.

1. *There exists a potential ψ (unique up to a constant) solving the weighted Poisson equation*

$$-\nabla\cdot\big(\rho\,(2D)\,\nabla\psi\big) = \delta\rho. \tag{6}$$

   Setting $u^\star := 2D\nabla\psi$ yields $u^\star \in \mathcal{U}_\rho(\delta\rho)$.

2. (**Orthogonal decomposition.**) *For any $u \in \mathcal{U}_\rho(\delta\rho)$, set $w := u - u^\star$. Then $\nabla\cdot(\rho w) = 0$ and*

$$\begin{aligned}\mathcal{E}_\rho(u) &= \mathcal{E}_\rho(u^\star) + \mathcal{E}_\rho(w),\\ \int \langle u^\star, D^{-1}w\rangle\, \rho\, dx &= 0.\end{aligned} \tag{7}$$

   Therefore, $u^\star$ is the unique minimiser of $\mathcal{E}_\rho$ over $\mathcal{U}_\rho(\delta\rho)$, determined by $(\rho, \delta\rho, D)$.

Proof sketch. Minimise $\mathcal{E}_\rho$ over $\mathcal{U}_\rho(\delta\rho)$ using a Lagrange multiplier ψ. The Euler–Lagrange condition yields $u^\star = 2D\nabla\psi$ and the weighted Poisson equation (6). Writing any feasible $u = u^\star + w$ with $\nabla\cdot(\rho w) = 0$ gives (7); see Appendix K.

Remark. Theorem 1 is an identifiability statement: given $(\rho, \delta\rho, D)$, all drifts producing the same marginal tangent form an affine gauge class, and the canonical representative is the unique gradient field with minimal kinetic energy. In smoothing or bridging objectives based on path-space KL, this is not an aesthetic choice: the KL kinetic term *is* the energy being minimized, so the posterior/control solution canonically removes any ρ–divergence-free "gauge" component segment-wise.

3.2. Posterior as a Schrödinger–Doob Transform

With the observational structure and per-time tilts now defined, we record the following existence result for the posterior and its induced marginals (see Lemma 3 in Appendix I).

Proposition 1 (Posterior Schrödinger–Doob representation)**.** *Under Assumption 1, let P_{post} be defined by (2). Then for any $Q \ll P_{ref}$,*

$$\mathcal{F}[Q] = D_{\mathrm{KL}}\big(Q\|P_{post}\big) - \log Z, \tag{8}$$

and hence $\arg\min_Q \mathcal{F}[Q] = P_{post}$ uniquely (a Schrödinger–Doob transform with fixed potentials). Moreover, letting $\mu_k := (P_{post})_{t_k}$, one has the endogenous-anchor entropy projection identity

$$P_{post} = \arg\min_{Q \ll P_{ref}:\ Q_{t_k} = \mu_k\ \forall k=0,\dots,K} D_{\mathrm{KL}}\big(Q\|P_{ref}\big). \tag{9}$$

Proof sketch. By Lemma 3, the anchors $\mu_k = (P_{\mathrm{post}})_{t_k}$ are well defined and the constrained entropy projection is unique, which yields identity (9). Details in Appendix I.

Remark. The identities in Proposition 1 are exact over the full admissible class $Q \ll P_{\mathrm{ref}}$. The "anchors" in (9) are *endogenous* posterior marginals $\mu_k = (P_{\mathrm{post}})_{t_k}$ (including μ_0, which is generally updated by future evidence), not the per-time Bayes tilts $\rho^{\mathrm{obs}}_{t_k}$ (unless $K = 1$); see Proposition 3. Hard-anchor bridges with prescribed *exogenous* $\{\nu_k\}$ are used only as controlled baselines, and the posterior itself can be viewed as the corresponding bridge when $\nu_k = \mu_k$. Crucially, (9) is not a re-definition: it is the bridge-to-posterior transfer that lets us import the hard-anchor Schrödinger bridge optimality system and geometry segment-wise to smoothing. In particular, combining Proposition 1 with Theorem 1 identifies the posterior/control drift between observations as the canonical minimal-energy representative within its gauge class.

3.3. Diffusion-Induced Geometry: Metric–Drift Separation

We now expose the geometric structure of the kinetic term $D_{\mathrm{KL}}(Q\|P_{\mathrm{ref}})$. For the posterior, this term induces the Onsager–Fokker metric with moving-frame kinematics (Definition 2); the posterior inherits a segment-wise geodesic structure via its endogenous anchors (Corollary 1). Diffusion-scale limits are treated in Sec. 4.2.

Definition 2 (Control-induced tangent / moving-frame derivative)**.** Let b and $D \succ 0$ define the reference Fokker–Planck operator $\mathcal{L}^*\rho := -\nabla\cdot(\rho b) + \nabla\cdot(D\nabla\rho)$. For a

marginal curve (ρ_t) we define its *control-induced tangent* (the derivative in the moving frame of the reference diffusion) by

$$\mathcal{D}_t\rho_t \;:=\; \partial_t\rho_t - \mathcal{L}^*\rho_t \;=\; \partial_t\rho_t + \nabla\cdot(\rho_t b) - \nabla\cdot(D\nabla\rho_t). \tag{10}$$

If (ρ_t, v_t) satisfies the controlled Fokker–Planck equation $\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = \nabla\cdot(D\nabla\rho_t)$, then $\mathcal{D}_t\rho_t = -\nabla\cdot(\rho_t(v_t - b))$ depends only on the control $u_t := v_t - b$. In particular, in the driftless *transport* regime $\partial_t\rho + \nabla\cdot(\rho v) = 0$ (no reference drift in the kinematics), the tangent is just $\mathcal{D}_t\rho = \partial_t\rho$ and g^D reduces to the standard Otto metric when $2D = I$.

Proposition 2 (Elliptic diffusion geometry and metric–drift separation (hard anchors))**.** *(Standard; see, e.g., Léonard (2014); Mikami & Thieullen (2006) for Schrödinger bridge optimality systems and Jordan et al. (1998); Benamou & Brenier (2000); Ambrosio et al. (2008) for the induced kinetic geometry.) Consider a reference diffusion* $\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t$ *with constant* Σ *(hence* $D := \frac{1}{2}\Sigma\Sigma^\top \succ 0$*) and drift* b *that is* C^1 *in* x *with bounded Jacobian uniformly in* t *(equivalently, globally Lipschitz in* x *uniformly in* t*). Fix times* $0 = t_0 < t_1 < \cdots < t_K \le T$ *and exogenous anchor marginals* $\{\nu_k\}_{k=0}^K$ *with* $\nu_0 \ll \rho_0$*, and let* Q^* *solve the multi-marginal Schrödinger bridge problem under these anchors. Then, on each open interval* (t_{k-1}, t_k)*, the marginal curve* (ρ_t) *of* Q^* *is a constant-speed geodesic for the metric* g^D with tangent identified as $\mathcal{D}_t\rho_t$ *in the sense of Definition 2, i.e. it minimises* $\int \|\mathcal{D}_t\rho_t\|_{g^D}^2\,\mathrm{d}t$ *among curves with fixed endpoints. The Onsager operator and metric are*

$$\begin{aligned} K_\rho^D\phi &:= -\nabla\cdot\big(\rho\,(2D)\,\nabla\phi\big), \\ g_\rho^D(\delta\rho_1, \delta\rho_2) &:= \int \langle\nabla\phi_1, (2D)\nabla\phi_2\rangle\,\rho\,\mathrm{d}x, \end{aligned} \tag{11}$$

with $K_\rho^D\phi_i = \delta\rho_i$ *(i.e.* $-\nabla\cdot(\rho(2D)\nabla\phi_i) = \delta\rho_i$*). The first-order optimality system reads*

$$\begin{aligned} &v_t - b = 2D\,\nabla\psi_t, \\ &-\partial_t\psi_t = \tfrac{1}{2}\|\nabla\psi_t\|_{2D}^2 + \langle\nabla\psi_t, b\rangle + \nabla\cdot\big(D\,\nabla\psi_t\big). \end{aligned} \tag{12}$$

Moreover, length–energy equivalence holds:

$$\begin{aligned} D_{\mathrm{KL}}(Q^*\|P_{\mathrm{ref}}) &= D_{\mathrm{KL}}(\nu_0\|\rho_0) \\ &\quad + \int_0^T\!\!\int \langle\nabla\psi_t, D\nabla\psi_t\rangle\,\rho_t\,\mathrm{d}x\mathrm{d}t. \end{aligned} \tag{13}$$

where $\mathcal{D}_t\rho_t$ *is the control-induced tangent from Definition 2. In particular, the metric tensor depends only on* D *(diffusion) and is independent of* b *(drift), which enters only via FP and HJB constraints.*

Remark. (Otto's W_2 geodesics.) The term $\langle\nabla\psi_t, b\rangle$ in (12) does not contradict Otto's W_2 geodesics: our kinematics is not the bare continuity equation $\partial_t\rho + \nabla\cdot(\rho v) = 0$. Rather, we measure motion in the moving frame of the reference diffusion via $\mathcal{D}_t\rho = \partial_t\rho - \mathcal{L}^*\rho$. In the control coordinates $u_t := v_t - b$ one has $\mathcal{D}_t\rho = -\nabla\cdot(\rho u_t)$ and the metric g^D depends only on D; the drift enters only through the reference generator $\mathcal{L}^*$ (i.e. the choice of moving frame). *Geometrically*, g^D plays the role of the Otto tensor, while $\mathcal{D}_t$ is the associated covariant derivative determined by the reference diffusion; this is why the geodesic equation acquires b-dependent terms when written in the laboratory frame.

Proof sketch. Apply Girsanov with fixed diffusion to obtain $D_{\mathrm{KL}}(Q\|P_{\mathrm{ref}}) = D_{\mathrm{KL}}(\nu_0\|\rho_0) + \frac{1}{4}\mathbb{E}_Q\int u^\top D^{-1}u$ for control drift $u = v - b$; introduce the Lagrangian with FP constraint $\partial_t\rho + \nabla\cdot(\rho v) = \nabla\cdot(D\nabla\rho)$ and vary in (v, ρ) to get (12). Define K_ρ^D by duality to recover (11) and the length–energy identity; details parallel Appendix K. In the Bayesian posterior case with discrete likelihood potentials, the same metric g^D governs the kinetic term, and the HJB equation holds on each (t_{k-1}, t_k) with jump conditions at observation times given by (34).

3.4. Anchor compatibility: posterior μ_k vs. per-time tilts ρ^{obs}

The per-time Bayes tilts $\rho_{t_k}^{\mathrm{obs}}$ are local evidence updates; the true posterior marginals μ_k coincide with them only when there is no cross-time information flow at t_k. The following proposition makes this precise via a single conditional factor.

Proposition 3 (Anchor compatibility criterion)**.** *Under Assumption 1 and any Markov reference* P_{ref} *with strictly positive transition density (in particular the OU prior of Sec. 2.2), the posterior marginal at* t_k *admits the disintegration*

$$\begin{aligned} \mu_k(\mathrm{d}x) &\propto r(t_k, x)\,\ell(y_k \mid x)\,h_k(x)\,\mathrm{d}x, \\ h_k(x) &:= E_{P_{\mathrm{ref}}}\big[\,L_{-k}(X)\,\big|\,X_{t_k}{=}x\big], \\ L_{-k}(X) &:= \prod_{j\neq k}\ell(y_j \mid X_{t_j}). \end{aligned} \tag{14}$$

where h_k *is the cross-time factor. Consequently,* $\mu_k = \rho_{t_k}^{\mathrm{obs}}$ *if and only if* h_k *is constant in* x*, i.e.*

$$h_k(x) \equiv C_k(y_{-k}) \quad (x\text{-independent}). \tag{15}$$

Equivalently (diffusion/HJB setting), in the Doob–h representation the time-t_k *potential coincides (up to an additive constant) with* $V_k(x) = -\log\ell(y_k \mid x)$.

Proof sketch. Disintegrate P_{post} w.r.t. P_{ref} and condition on $X_{t_k} = x$; the Radon–Nikodym derivative factorises as in (14) using only Markovity and domination. Constancy of the conditional factor is necessary and sufficient for cancellation. In the diffusion/HJB formulation, the Doob–h equivalence follows from the jump condition for the HJB value at t_k.

Remark. A per-time compatibility score and robustness remarks for *approximate* anchors are provided in Appendix D–E.

4. Corollaries and Extensions

4.1. Posterior Geodesic Structure and OU Specialization

Corollary 1 (Posterior as an endogenous-anchor bridge (segment-wise geodesic structure))**.** *Under Assumption 1, let P_{post} be the soft-potential posterior (2) with marginals μ_t and $\mu_k := \mu_{t_k}$. Assume the reference diffusion has constant Σ with $D \succ 0$ and drift b that is C^1 in x with bounded Jacobian uniformly in t, and that the endogenous-anchor projection in (9) is feasible and unique (Lemma 3). Then the posterior is the unique hard-anchor bridge with anchors $\{\mu_k\}$, and on each open interval (t_{k-1}, t_k) its marginal curve (μ_t) is a constant-speed geodesic for g^D with tangent identified as $\mathcal{D}_t\mu_t$ (Definition 2). Moreover, there exists a potential ψ such that on each segment (t_{k-1}, t_k) one has the optimality system (12) together with the FP constraint, and at observation times the jump conditions (34) hold.*

Proof sketch. By Proposition 1, P_{post} is exactly the minimiser of $D_{\text{KL}}(\cdot\|P_{\text{ref}})$ under the hard marginal constraints $Q_{t_k} = \mu_k$ for $k = 0, \dots, K$, hence Proposition 2 applies with $\nu_k = \mu_k$. The jump conditions follow from the soft-potential term (Appendix K, (34)).

Corollary 2 (Isotropic OU case)**.** *For $D = \sigma^2 I/2$ (Ornstein–Uhlenbeck), the metric reduces to*

$$g_\rho^{\text{OF}}(\delta\rho_1, \delta\rho_2) = \frac{1}{\sigma^2}\int_{\mathbb{R}^d} \nabla\phi_1 \cdot \nabla\phi_2 \, \rho \, dx, \qquad (16)$$

and the kinetic energy is exactly the Benamou–Brenier action of the moving-frame tangent $\mathcal{D}_t\rho_t$ [2].

Auxiliary baseline (fixed anchors). As a controlled baseline used only for geometry ablations, fixed-anchor multi-marginal Schrödinger bridges are Onsager–Fokker geodesics under OU dynamics.

Corollary 3 (Onsager–Fokker geometry of fixed-anchor Schrödinger bridges, OU case)**.** *Fix times $0 = t_0 < t_1 < \cdots < t_K \le T$ and exogenous anchors $\{\nu_k\}_{k=0}^K$ with $\nu_0 = \rho_0$. Let the OU reference process P_{ref} have diffusion coefficient $\sigma > 0$ and a continuously differentiable drift $b = -Ax$. Assume the multi-time Schrödinger system under anchors $\{\nu_k\}$ is feasible and admits a unique minimiser (Appendix J). Then the unique multi-marginal Schrödinger bridge Q^* satisfying $Q^*_{t_k} = \nu_k$ has time marginals $(\rho_t)_{t\in[0,T]}$ that form Onsager–Fokker geodesic segments on the manifold $(\mathcal{P}(\mathbb{R}^d), g^{\text{OF}})$ equipped with the diffusion-scaled Otto–Wasserstein tensor. On each interval (t_{k-1}, t_k) the evolution is governed by the Hamilton–Jacobi–Bellman (HJB) equation for the scalar potential ψ together with the controlled Fokker–Planck (FP) constraint (see (33), (30)):*

$$\begin{aligned} v_t - b &= \sigma^2 \nabla\psi_t, \\ -\partial_t\psi_t &= \tfrac{\sigma^2}{2}\|\nabla\psi_t\|^2 + \langle\nabla\psi_t, b(x)\rangle + \tfrac{\sigma^2}{2}\Delta\psi_t \end{aligned} \qquad (17)$$

Here ψ is the scalar potential from Proposition 2 specialised to $D = \sigma^2 I/2$ (so $v - b = 2D\nabla\psi$), and in the Doob–h representation one has $\rho_t \propto f_t\, g_t\, r(t,\cdot)$ for Schrödinger potentials f_t, g_t.

In particular, the induced metric satisfies

$$\|\mathcal{D}_t\rho_t\|^2_{g^{\text{OF}}(\rho_t)} = \sigma^{-2}\int_{\mathbb{R}^d} \|v_t - b\|^2 \, \rho_t \, dx,$$

where $\mathcal{D}_t\rho_t = \partial_t\rho_t + \nabla\cdot(\rho_t b) - \frac{\sigma^2}{2}\Delta\rho_t$.

Proof sketch. Directly from Proposition 2 with $D = \sigma^2 I/2$; full details are provided in Appendix K.

4.2. Diffusion-Scale Limits of the Soft-Potential Posterior

We vary only the isotropic diffusion scale σ of the reference dynamics while keeping the observation potentials $V_k = -\log \ell(y_k \mid \cdot)$ fixed. For each $\sigma > 0$ let P^σ_{ref} be the law of the reference diffusion

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim \rho_0,$$

and define the soft-potential posterior

$$\frac{\mathrm{d}P^\sigma_{\text{post}}}{\mathrm{d}P^\sigma_{\text{ref}}} = Z_\sigma^{-1}\exp\Big(-\sum_{k=1}^K V_k(X_{t_k})\Big). \qquad (18)$$

Write $\mu^\sigma_t := (P^\sigma_{\text{post}})_t$ and $\mu^\sigma_k := \mu^\sigma_{t_k}$. Since $D_{\text{KL}}(Q\|P^\sigma_{\text{ref}}) = D_{\text{KL}}(Q_0\|\rho_0) + \frac{1}{2\sigma^2}\mathbb{E}_Q[\int \|u_t\|^2\,\mathrm{d}t]$, the inverse temperature $\beta := \sigma^{-2}$ multiplies the kinetic-energy term (up to the σ-independent boundary term $D_{\text{KL}}(Q_0\|\rho_0)$) while the potentials V_k are fixed. We characterise the endpoint regime $\sigma \to 0$ and, *only under asymptotic mixing of the reference kernels* (finite grids/compact domains with uniformly positive kernels; not $\mathbb{R}^d$), the regime $\sigma \to \infty$ for the *posterior* family P^σ_{post}.

[2] The tensor g^{OF} is the diffusion-scaled Otto metric, while the kinematics uses the control-induced tangent $\mathcal{D}_t\rho_t = \partial_t\rho_t - \mathcal{L}^*\rho_t$ (Definition 2); full details are in Appendix K.

4.2.1. Low diffusion ($\sigma \to 0$): deterministic-flow posterior

Let $\Phi_t : \mathbb{R}^d \to \mathbb{R}^d$ be the flow of the ODE $\dot{x} = b(t, x)$. Define the deterministic posterior on initial conditions by

$$\mu_0^0(\mathrm{d}x_0) := \bar{Z}^{-1}\, \rho_0(x_0) \exp\Big(-\sum_{k=1}^{K} V_k(\Phi_{t_k}(x_0))\Big)\, \mathrm{d}x_0,$$
$$\bar{Z} := \int_{\mathbb{R}^d} \rho_0(x_0) \exp\Big(-\sum_{k=1}^{K} V_k(\Phi_{t_k}(x_0))\Big)\, \mathrm{d}x_0. \tag{19}$$

and let P^0_{post} be the law on Ω of the deterministic path $X_t = \Phi_t(X_0)$ with $X_0 \sim \mu_0^0$. **Limit as $\sigma \to 0$ (informal).** Under standard small-noise assumptions, the posterior concentrates on deterministic prior trajectories: $P^\sigma_{\text{post}} \Rightarrow P^0_{\text{post}}$ and hence $\mu_t^\sigma \Rightarrow (\Phi_t)_\# \mu_0^0$. This is a classical small-noise regime for SDE posteriors; see, e.g., Freidlin & Wentzell (1998); Fleming & Soner (2006) and the self-contained statement in Appendix L (Proposition 9).

4.2.2. High diffusion ($\sigma \to \infty$): time-slice decoupling

High diffusion suppresses the kinetic penalty, but a clean $\sigma \to \infty$ decoupling statement requires a *uniform* mixing regime (Doeblin/Dobrushin contraction of the reference kernels), so we state the limit theorem only for finite grids or compact domains with uniformly positive kernels. On $\mathbb{R}^d$ this fails even for OU/Gaussian kernels: for any $\sigma > 0$, $\sup_{x,x'} \|K_k^\sigma(x,\cdot) - K_k^\sigma(x',\cdot)\|_{\mathrm{TV}} = 1$ (take $\|x - x'\| \gg \sigma$), hence we make no unconditional total-variation time-slice decoupling claim on $\mathbb{R}^d$. For each $\sigma > 0$ define the per-time evidence tilt of the reference marginal

$$\rho_{t_k}^{\mathrm{obs},\sigma}(x) := \frac{r^\sigma(t_k, x)\, e^{-V_k(x)}}{Z_{k,\sigma}},$$
$$Z_{k,\sigma} := \int_{\mathbb{R}^d} r^\sigma(t_k, x)\, e^{-V_k(x)}\, \mathrm{d}x. \tag{20}$$

where $r^\sigma(t_k, \cdot)$ is the time-t_k marginal density of P^σ_{ref}. **Limit as $\sigma \to \infty$ under mixing (informal).** On finite grids or compact domains where the reference kernels become uniformly mixing as $\sigma \to \infty$, posterior slices converge to the local evidence tilts $\rho_{t_k}^{\mathrm{obs},\sigma}$ in total variation and cross-time dependence vanishes (mutual information $\to 0$). A self-contained statement and assumptions appear in Appendix L (Proposition 10); a discrete-grid proof is given in Appendix M. **Optional: rates.** A TV rate bound in terms of $\mathrm{osc}(\log h_k^\sigma)$ is stated and proved in Appendix M (Lemma 8).

Corollary 4 (Thermodynamic scaling and a mutual-information order parameter)**.** *Assume isotropic diffusion $\Sigma = \sigma I$ so that $D = \sigma^2 I/2$ and set $\beta := \sigma^{-2}$. For the posterior P^σ_{post}, let $\pi_k^\sigma := (P^\sigma_{\text{post}})_{t_{k-1},t_k}$ and define*

$$\mathcal{I}_k(\sigma) := D_{\mathrm{KL}}\big(\pi_k^\sigma \,\|\, \mu_{k-1}^\sigma \otimes \mu_k^\sigma\big) = I_{P^\sigma_{\text{post}}}(X_{t_{k-1}}; X_{t_k}).$$

Then β multiplies the kinetic energy while the potentials are fixed, so varying σ is a thermodynamic homotopy. Moreover:

1. ***(High temperature / decoupling.)** Under the assumptions of Proposition 10, one has $\mathcal{I}_k(\sigma) \to 0$ as $\sigma \to \infty$.*
2. ***(Low temperature / deterministic coupling.)** Under the assumptions of Proposition 9, the limiting posterior is supported on the deterministic flow $X_{t_k} = \Phi_{t_k}(X_0)$. On continuous state spaces this implies $\mathcal{I}_k(\sigma) \to +\infty$ as $\sigma \to 0$ (the limit coupling is singular with respect to $\mu_{k-1}^0 \otimes \mu_k^0$), while on a finite grid $\mathcal{I}_k(\sigma)$ converges to the mutual information of the deterministic limit transition.*

4.3. Extensions: Open Systems via Reaction–Transport Surrogates

When mass is not conserved, one can replace the closed-system posterior line with a reaction–transport surrogate, which augments transport with a Fisher–Rao-type reaction term. We include an illustrative derivation, static–dynamic equivalence, and diagnostics in Appendix F–H.

5. Applications to a High-dimensional Neural Example

5.1. Implementation Notes: Grid Posterior and Controlled Baselines

Exact posterior on grids (Doob–h / forward–backward). On a discretised grid, the posterior over time-slices $x_{0:K}$ has the Markov factorisation

$$P_{\text{post}}(x_{0:K}) \propto r(t_0, x_0) \prod_{k=1}^{K} K_k(x_{k-1}, x_k)\, \ell(y_k \mid x_k),$$

where $K_k := K_{t_{k-1} \to t_k}$. Posterior marginals μ_k are computed by standard message passing; for instance, an unnormalised forward recursion is

$$\alpha_k(x_k) \propto \ell(y_k \mid x_k) \int \alpha_{k-1}(x_{k-1})\, K_k(x_{k-1}, x_k)\, \mathrm{d}x_{k-1},$$

with integrals implemented by quadrature and stabilised in the log domain (Appendix N). On an N-point grid this costs $O(KN^2)$ time and $O(KN)$ memory.

Exogenous-anchor bridges (controlled baseline). For prescribed anchors $\{\nu_k\}$, we solve the multi-marginal Schrödinger bridge via log-domain IPFP/Sinkhorn scaling (Csiszár, 1975; Franklin & Lorenz, 1989; Lemmens & Nussbaum, 2012); see Appendix N and Appendix O.

5.2. High-dimensional neural gauge canonicalization

For the synthetic LGSSM–OU compute-limited vs. long-run dashboard, see Appendix C.1 (Figure 4). Grid-level diagnostics and recipe checks are in Appendix C (Figure 3).

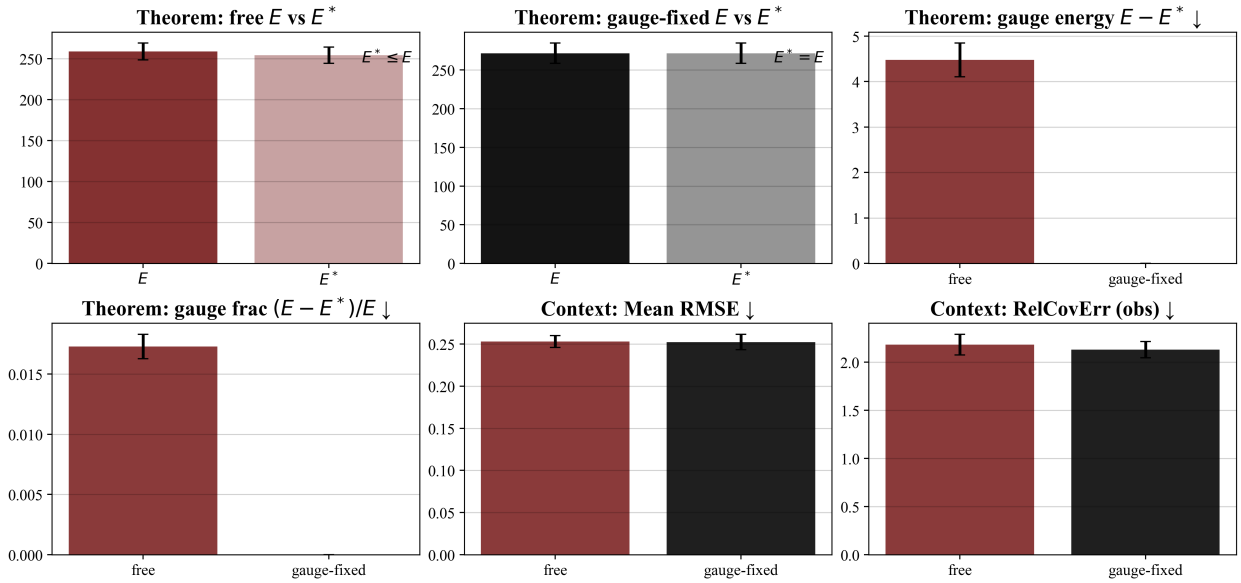

Figure 2. CMU Mocap observations (Carnegie Mellon University, 2002) (mean±sd over 5 seeds). Panels (a–d): theorem-aligned gauge decomposition at fixed RTS marginals ρ_k, where total energy E decomposes into canonical energy $E^\star$ plus gauge remainder $E - E^\star \geq 0$. The free parameterization exhibits a stable nonzero gauge component (4.476 ± 0.371, $1.73\% \pm 0.10\%$), while the canonical (potential/gradient) parameterization has zero gauge remainder by construction. Panels (e–f): context-only metrics (Mean RMSE and RelCovErr (obs)), included only for comparison and not part of the theorem statement.

Canonical parameterization (practical). In high dimensions, rather than solving the weighted Poisson problem (6) explicitly, we parameterize a scalar potential $\psi_\theta(t, x)$ and set $u_\theta(t, x) := 2D\, \nabla_x \psi_\theta(t, x)$, enforcing the canonical (gradient) drift class by construction.

CMU Mocap observations: theorem-aligned energy comparison. The theorem statement is: at a fixed marginal ρ (and fixed diffusion geometry), gauge canonicalization selects the unique same-divergence gradient representative with minimal kinetic energy (Theorem 1). We therefore fix the RTS posterior marginals ρ_k (Gaussian, given the same $y_{1:K}$) and decompose the learned linear drift energy into its canonical (same-divergence) gradient representative plus a nonnegative gauge remainder. Figure 2 makes the theorem-aligned statement explicit: at fixed RTS marginals ρ_k, canonicalization removes a nonnegative gauge remainder (gauge energy 4.476 ± 0.371, fraction $1.73\% \pm 0.10\%$).

6. Conclusions

Conclusion. Diffusion D fixes the Onsager–Fokker geometry of smoothing, while drift is identifiable only up to a ρ–divergence-free gauge; KL selects a canonical minimal-energy representative (Theorem 1). With isotropic diffusion, σ is a temperature parameter governing cross-time coupling, quantified by $\mathcal{I}_k(\sigma)$ (Corollary 4). Implementation code is provided in the supplementary material.

Practical recipe. Choose geometry (scale/anisotropy of D) before tuning dynamics; canonicalize drift via control energy/path KL; use per-time tilts only as local baselines (or exogenous anchors) and report compatibility residuals (Proposition 3); validate with temperature sweeps and log-domain stability diagnostics. For neural SDE smoothers, parameterize the control drift as $u_\theta(t, x) = 2D\, \nabla_x \psi_\theta(t, x)$ (scalar potential network) and keep the kinetic-energy term in (5) explicit in the loss; if learning an unconstrained vector drift, add an auxiliary potential head ψ_ϕ and penalize deviations from canonical form via $\mathbb{E}\|u_\theta - 2D\nabla_x \psi_\phi\|^2$.

Extensions. For hypoelliptic references (e.g., kinetic Langevin), the induced continuum geometry is sub–Riemannian and extending the metric–drift separation requires new tangent/energy characterisations. When mass is not conserved, one can replace the closed-system posterior line with a reaction–transport surrogate (HK/WFR); we include an illustrative derivation and diagnostics in Appendix F–H.

Impact Statement

Neural SDE pipelines often treat diffusion and drift as interchangeable optimisation knobs; this can yield models that are statistically plausible yet physically incoherent, and trajectories whose intermediate-time dynamics are not supported by data. We separate diffusion (which fixes geometry/transport cost) from drift (identified from marginals only up to a ρ-divergence-free gauge) and provide a canonical minimal-energy representative selected by the Schrödinger–

KL objective. When diffusion is constrained by domain knowledge (e.g., known noise scales or conservation laws), the gauge-canonicalization principle suppresses spurious, overconfident drift components that do not affect marginals but inflate control energy; this improves interpretability and reduces the risk of "hallucinated" dynamics in safety-critical interpolation and smoothing.

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Appendix

Appendix roadmap. Related work: Appendix A. Experiments and diagnostics: Appendix B–C. Algorithms and implementation details: Appendix N–O. Proofs: Appendix I–M (including Appendix K and Appendix L). Optional extensions (HK/WFR): Appendix F–H.

A. Related Work

A.1. A. Bayesian smoothing and state-space inference

Classical Bayesian smoothing in linear–Gaussian state-space models is solved exactly by Kalman filtering and RTS smoothing (Kalman, 1960; Rauch et al., 1965; Särkkä, 2013). For nonlinear models, common approximations include Gaussian closures (e.g. iterated extended/unscented smoothers (Bell, 1994; Särkkä, 2008)) and variational or particle-based families (Blei et al., 2017; Jordan et al., 1999; Naesseth et al., 2018). Our paper does not propose a new general-purpose smoother in this taxonomy. Rather, we fix a dominated reference diffusion and treat smoothing as an *exact posterior path measure* defined by a Gibbs/Doob tilt (Sec. 2.3–3.2), then expose the *geometry* induced by the KL kinetic term and derive diagnostics that quantify when common per-time approximations (local evidence tilts) are inadequate (Proposition 3 and Appendix D).

A.2. B. Doob h-transforms, Schrödinger bridges, and IPFP/Sinkhorn

The representation of tilted path measures as Doob h-transforms is classical; in continuous time this connects to stochastic control and Föllmer-type drifts (Föllmer, 1988). The Schrödinger bridge problem—entropy projection of a reference path measure subject to marginal constraints—originates with Schrödinger and is extensively surveyed by Léonard (Léonard, 2014), with dynamical OT/control formulations in, e.g., Mikami & Thieullen (2006). In discrete settings, iterative proportional fitting / Sinkhorn-type scaling is a standard solver (Csiszár, 1975; Franklin & Lorenz, 1989). Recent machine-learning work connects Schrödinger bridges to diffusion/score-based estimation and neural parameterisations (De Bortoli et al., 2021; Vargas et al., 2021; Liu et al., 2022). In our paper, the soft-potential Bayesian posterior is exactly a Doob tilt of the reference diffusion (Proposition 1); grid inference reduces to standard forward–backward message passing (Appendix N). Multi-marginal Schrödinger bridges under *exogenous* anchors appear only as controlled baselines (Sec. 5.1), solved via IPFP on grids (Appendix N–O).

A.3. C. Otto calculus, Onsager–Fokker geometry, and moving-frame tangents

The Otto calculus viewpoint identifies a Riemannian structure on probability measures where (in the conservative transport case) tangent vectors are represented by gradient potentials and kinetic energy induces the W_2-geometry (Jordan et al., 1998; Otto, 2001; Benamou & Brenier, 2000; Villani, 2009; Ambrosio et al., 2008). Our setting is not the bare continuity equation: kinematics are measured in the *moving frame* of a reference diffusion. We therefore define the control-induced tangent $\mathcal{D}_t\rho_t := \partial_t\rho_t - \mathcal{L}^*\rho_t$ (Definition 2), and show that for fixed diffusion D the induced metric tensor g^D depends only on D, while the drift b appears only through FP/HJB constraints and the moving-frame operator (Proposition 2 and Remark following it). This separates "geometry" (metric tensor) from "dynamics" (reference generator / admissibility), which is frequently conflated in applied SDE training narratives.

A.4. D. Drift non-identifiability (gauge) and minimal-energy selection

In Wasserstein/Otto geometry, tangents are equivalence classes of velocity fields modulo divergence-free directions; the gradient representative is the minimal kinetic-energy element within the class (Otto, 2001; Ambrosio et al., 2008; Benamou & Brenier, 2000). Our paper adopts the same structural principle but in a diffusion-controlled (Onsager–Fokker) setting: with fixed D, many drifts yield the same marginal tangent because only the weighted divergence $\nabla\cdot(\rho u)$ is observable from marginals. We formalize this as a ρ-weighted gauge equivalence (Definition 1) and show that the Schrödinger–KL objective selects a unique canonical representative by minimizing kinetic energy, yielding a weighted Poisson equation and an orthogonal decomposition (Theorem 1). This makes precise which parts of drift are identifiable from marginal evolution and clarifies what "gauge fixing" should mean in smoothing/bridge objectives based on path-space KL.

A.5. E. Score-based diffusion models (fixed diffusion schedules) and our scope

In many score-based diffusion models, the forward diffusion schedule (hence the metric tensor induced by $D(t)$) is fixed a priori (Sohl-Dickstein et al., 2015; Song & Ermon, 2019; Ho et al., 2020; Song et al., 2021). Our main claims are about Bayesian *smoothing* under a chosen reference diffusion: (i) changing D changes the geometry (transport cost), and (ii) even at fixed D the drift is only identifiable up to weighted divergence-free gauge directions when reasoning from time-marginals. In fixed-diffusion generative modeling, the primary objective is often to match marginals/endpoints, so gauge directions may be largely irrelevant if one does not interpret intermediate trajectories. In contrast, for smoothing/bridge interpretations where intermediate-time paths carry semantic meaning (e.g., interpolation under a physical prior), our "geometry-first, gauge-fixed" viewpoint explains why trajectories can be geometrically incoherent if geometry (D) and dynamics (b) are treated as interchangeable knobs.

A.6. F. Temperature limits, small noise, and high noise under mixing

Small-noise limits connect SDE posteriors to deterministic flows and large-deviation/control principles (Freidlin & Wentzell, 1998; Fleming & Soner, 2006). Our "temperature" parameterization $\beta = \sigma^{-2}$ makes explicit that varying σ rescales kinetic energy while keeping observation potentials fixed (Sec. 4.2). For large noise, we prove time-slice decoupling only under mixing regimes (finite grids or compact domains with uniformly positive kernels; Proposition 10), aligning the proof technique with classical Doeblin/Dobrushin contraction ideas and Hilbert-metric arguments used in IPFP convergence analyses (Birkhoff, 1957; Lemmens & Nussbaum, 2012; Carlier et al., 2023). On $\mathbb{R}^d$, uniform total-variation mixing generally fails; we therefore separate what is proved (mixing regimes) from what is heuristic or purely empirical in discretized diagnostics (Sec. 3.4 and Sec. 4.2).

B. Experiments and Diagnostics: Details

B.1. General Implementation Details

All floating-point operations were performed using 64-bit precision to ensure numerical accuracy. The specifics of the IPFP implementation, including the use of log-space computations, Anderson acceleration, and adaptive ε-scaling, are detailed in Appendix N.

B.2. OU–GMM Benchmark: Setup

Anchors and prior. Anchors are 2D Gaussian mixtures constructed to be consistent with a separable OU prior: means relax exponentially toward equilibrium and variances interpolate to the OU equilibrium values along each axis. We use $K = 8$ time points over $[0, 2]$; each marginal is a 3-component mixture with weights $[0.25, 0.5, 0.25]$, time-varying peak separations (kept $> 2\sigma$), and mild cross-axis shifts to maintain distinct multi-modality. For baselines that require observations, per-time point clouds are drawn from the target marginals on the same grid.

MMSB SMOOTHER configuration.

- **Grid/quadrature:** 2D uniform grid with 128×128 nodes; per-axis bounds $\pm 5\,\sigma_{eq}$ with $\sigma_{eq}^2 = \sigma^2/(2\theta)$; 2D trapezoidal weights.
- **Reference process:** Separable OU with axis-wise parameters (example): $\theta_x = 1.5$, $\sigma_x = 0.4$ and $\theta_y = 2.0$, $\sigma_y = 0.3$, equilibrium means 0.
- **Anchors (Targets):** $K = 8$ OU-consistent 2D GMMs over $[0, 2]$; weights $[0.25, 0.5, 0.25]$; initial scales are fractions of σ_{eq} and then evolved by OU. Peak separations kept $> 2\sigma$.

Baselines.

- **RTS/IEKS/UKS:** Gaussian smoothers under the same OU prior and grid: linear Rauch–Tung–Striebel (RTS; Rauch et al., 1965), iterated-extended (IEKS; Bell, 1994; Särkkä, 2013), and unscented (UKS; Särkkä, 2008). Likelihoods and L^1 metrics use the same trapezoidal quadrature as MMSB SMOOTHER.
- **SVI:** Factorized Gaussian smoother $q(\mathbf{X}) = \prod_t \mathcal{N}(x_t; \mu_t, \operatorname{diag} \sigma_t^2)$ trained by reparameterized ELBO.
- **SMC:** KDE-based likelihoods consistent with OU-GMM anchors; density-level metrics (cross-entropy/NLL, HPD coverage/area) evaluated on the same 128×128 grid. Particle counts and backward-sampling draws tuned for stability

with ensemble averaging.

- **Point-observation protocol:** When needed, at each time we sample $n = 100$ points from the target marginal and use isotropic Gaussian noise with s.d. 0.2 of the empirical s.d.

OU–GMM diagnostic metrics. We use a 2D OU–GMM example to visualise geometry-sensitive behaviors. Table 1 reports density-level and shape-sensitive metrics to contextualise phenomena (mode preservation vs moment-matching). Baselines (RTS/IEKS/UKS; FFBSi/RBPS/PGAS) are included as references rather than competitors; this section does not make SOTA claims.

Table 1. OU–GMM diagnostic metrics (controlled diagnostic evaluation).

	Density-level		Shape				
Method	Density LL	Point Obs LL	CRPS ($\times 10^{-3}$)	W_2 ($\times 10^{-3}$)	SWD	HPD@90%	Runtime (s)
MMSB SMOOTHER	0.39	0.44	1.50	1.39	0.02	90.01	57.07
RTS	−21.39	−22.19	51.33	34.37	0.05	13.58	1.50
IEKS	−21.24	−22.02	50.00	34.12	0.05	13.94	1.50
SVI	−21.21	−20.51	52.00	35.72	0.05	29.83	152.10
UKS	−6.65	−6.93	14.20	8.21	0.20	80.29	1.20
FFBSi	−1.85	−0.45	14.80	3.70	0.14	59.54	13.60
RBPS	−2.95	−0.52	14.90	4.00	0.15	57.43	8.10
PGAS	−0.51	−0.52	13.40	4.90	0.15	63.83	745.70

Larger is better: Density LL(per time), HPD@90%, State-LL(per obs). Smaller is better: CRPS, W_2, SWD.
All metrics share the same 128×128 grid and trapezoidal quadrature (endpoints 1/2).

B.3. Ablation: prior-as-geometry (details)

We validate our central claim with a deliberately minimal ablation that changes *only* the reference measure while keeping data, anchors, solver, and all hyperparameters fixed. Concretely, we use the 2D OU–GMM benchmark and solve the multi-marginal Schrödinger bridge (MMSB) with our log-domain IPFP (64-bit, compiled loop) on a fixed grid and $K{=}8$ time points. To expose geometry, we pin only the endpoints as anchors ($\{0, T\}$) and treat all intermediate times as *free*; performance is measured *exclusively* on these non-anchored *holdout* times. Primary metrics are density-level distances (mean L^1 and Hellinger) on holdout slices; sample-based distances (e.g., SWD, Sinkhorn) are used for sanity checks.

Hypothesis. Diffusion (σ) sets the metric (Onsager/Otto tensor $\propto \sigma^{-2}$); drift b affects FP/HJB constraints but not the metric.

Setup. OU–GMM (2D), fixed anchor marginals across all conditions; endpoints-only anchoring or sparse $K{=}8$ anchors; identical grids and seeds across runs. We compare five references: (A) *matched* OU (correct θ, μ, σ); (B) *drift-mismatch* (change mean reversion/equilibrium mean, correct σ); (C) *diffusion-mismatch* (scale σ, correct θ, μ); (D) *anisotropic diffusion* (fixed trace, $\sigma_y/\sigma_x{>}1$); (E) *near-Brownian drift* (very small θ, correct σ).

Metrics. Holdout L^1 and Hellinger distances (lower is better), reported only on non-anchored slices.

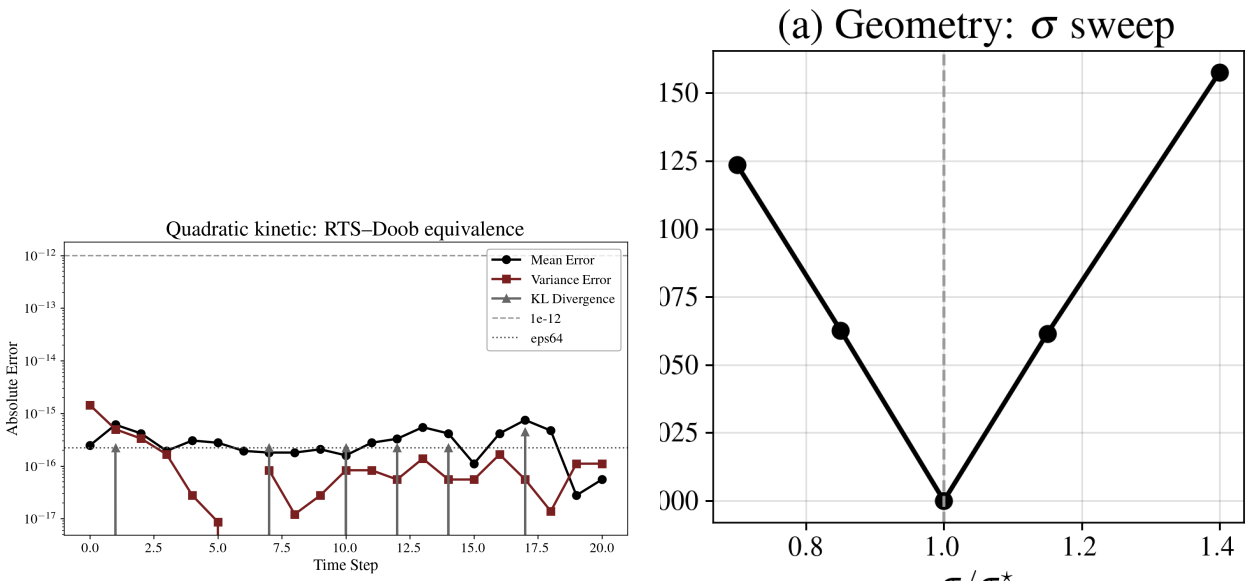
Findings. With endpoints-only anchoring, (C) *diffusion-mismatch* incurs far larger errors than (B) *drift-mismatch*. A representative run (endpoints anchors, $K{=}8$) shows mean holdout L^1: A 0.66, B 0.83, C 1.94; Hellinger: A 0.31, B 0.48, C 0.94. Thus $\Delta_{\text{C-A}} \gg \Delta_{\text{B-A}}$ in both metrics. Anisotropic diffusion (D) reveals *geometric bias*: changing anisotropy systematically deforms the interpolating path and can outperform an isotropic prior on certain slices under fixed anchors—consistent with diffusion defining the geometry. Drift-only changes (E) yield much flatter error curves.

These observations align with the theory: the Onsager/Otto tensor that governs geodesics scales as σ^{-2} and is independent of the drift b. Changing σ (even without touching θ, μ) changes the manifold and its geodesics; changing θ or μ at fixed σ preserves the metric and primarily affects constraints. Robustness checks: (i) *σ-scaling sweep* produces a (near-)monotone increase in holdout L^1/Hellinger as $\|\Sigma\|$ departs from the matched value; (ii) *anisotropy sweep* (fixed trace, varying σ_y/σ_x) deforms the interpolating path predictably; (iii) *drift sweep* yields much flatter curves than diffusion sweeps.

Conclusion. In MMSB bridges, diffusion sets the Riemannian geometry and the interpolating marginal curve; drift influences feasibility but not the metric. Practically, prioritise getting the diffusion (scale and anisotropy) right before tuning dynamics—especially when anchors are sparse and geometric interpolation dominates.

C. Experiments: Additional Results

Additional diagnostics. Figure 4 reports the neural gauge-fixing summary. Figure 3 collects grid-level diagnostics and geometry-first recipe checks; the remaining plots below provide deeper breakdowns for implementation sanity checks and anchor-compatibility.



(a) RTS ≡ MMSB SMOOTHER. *(b)* σ sweep (diffusion/temperature).

Figure 3. Grid-level diagnostics/recipe dashboard. (a) Linear–Gaussian sanity check: Doob–h posterior matches RTS to machine precision. (b) Diffusion-scale sensitivity (temperature view): varying σ changes the objective sharply, consistent with $\beta = \sigma^{-2}$ multiplying kinetic energy.

C.1. High-dimensional neural gauge fixing (compute-limited vs. long-run)

We report two complementary regimes of the high-dimensional LGSSM–OU neural gauge-fixing experiment (5 seeds: 0–4). In the *compute-limited* setting we cap training to a small fixed budget (and use a more under-observed sensing map), which makes the energy gap more visible at comparable error. In the *long-run* setting we train substantially longer with the default sensing dimension, where both parameterizations are pushed closer to the same KL/free-energy optimum and the energy gap shrinks. Metrics: RMSE(mean) compares the smoothed mean to RTS truth; $\mathrm{RelCovErr}_{\mathrm{obs}} := \frac{\mathbb{E}\|\widehat{\Sigma}_{HX} - \Sigma^*_{HX}\|_F}{\mathbb{E}\|\Sigma^*_{HX}\|_F}$ compares the projected covariance $\Sigma_{HX} = \mathrm{Cov}(HX_{t_k})$ in observation space; Energy is the kinetic term; Gauge ratio is a proxy for antisymmetric (gauge) content (zero by construction for gauge-fixed). Note that $\mathrm{RelCovErr}_{\mathrm{obs}}$ can exceed 1 since we do not train on covariance matching and $\|\Sigma^*_{HX}\|_F$ can be small when $m \ll d$; we use it as a strict second-order sanity check and interpret it primarily through relative comparisons across parameterizations. Summary bar plots are in Figure 4.

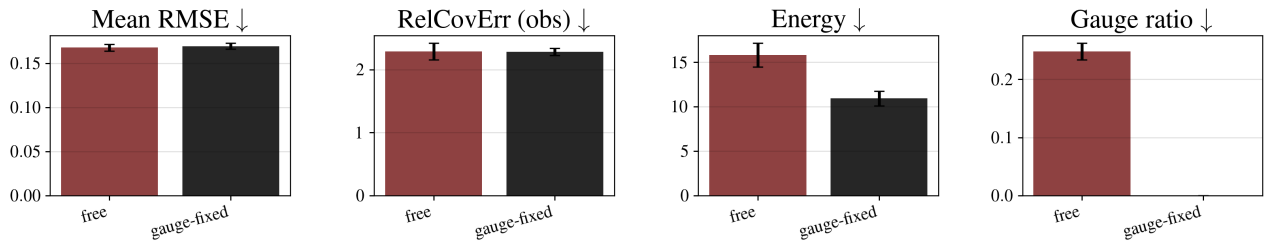
Compute-limited (fixed budget). Configuration: d=64, m=4, K=16, 50 training steps per method, seeds 0–4.

Table 2. Compute-limited neural gauge fixing (mean±sd over 5 seeds).

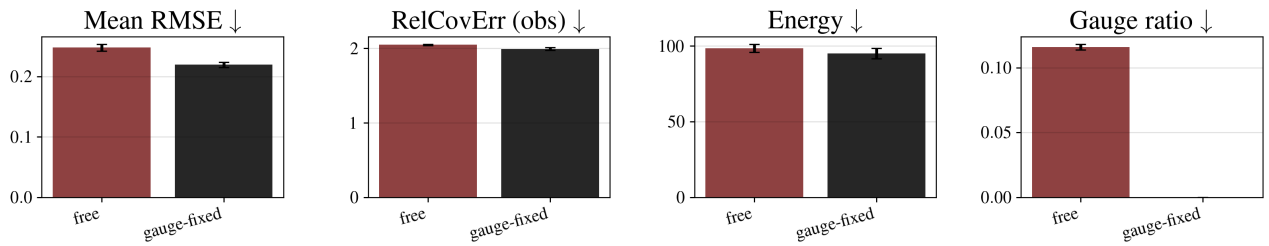
Method	RMSE(mean)↓	$\mathrm{RelCovErr}_{\mathrm{obs}}$ ↓	Energy↓	Gauge ratio↓
free	0.1679 ± 0.0037	2.2893 ± 0.1320	15.8287 ± 1.3436	0.2483 ± 0.0144
gauge-fixed	0.1697 ± 0.0035	2.2804 ± 0.0548	10.9359 ± 0.8404	0.0000 ± 0.0000

Long-run (near-converged). Configuration: d=64, m=16, K=16, 8000 training steps per method, seeds 0–4.

Figure 6 (Per-time diagnostics, Doob–h vs ρ^{obs}). We report a per-time breakdown of the anchor-compatibility diagnostics (Proposition 3); see the figure caption for panel definitions.



(a) Compute-limited setting: d=64, m=4, K=16, 50 training steps (mean±sd over 5 seeds).



(b) Long-run setting: d=64, m=16, K=16, 8000 training steps (mean±sd over 5 seeds).

Figure 4. High-dimensional neural gauge fixing under fixed diffusion geometry. Bars compare a free (vector) control drift parameterization versus a gauge-fixed (potential/gradient) parameterization. Gauge fixing removes antisymmetric (gauge) content by construction (gauge ratio ≈ 0).

Table 3. Long-run neural gauge fixing (mean±sd over 5 seeds).

Method	RMSE(mean)↓	$\text{RelCovErr}_{\text{obs}}$ ↓	Energy↓	Gauge ratio↓
free	0.2481 ± 0.0055	2.0454 ± 0.0078	98.5924 ± 2.6343	0.1159 ± 0.0021
gauge-fixed	0.2197 ± 0.0042	1.9905 ± 0.0182	95.1099 ± 3.3907	0.0000 ± 0.0000

D. Anchor-Compatibility Diagnostic

When $K > 1$, posterior anchors $\mu_k = (P_{\text{post}})_{t_k}$ generally differ from per-time tilts $\rho^{\text{obs}}_{t_k} \propto r(t_k, \cdot)\, \ell(y_k \mid \cdot)$. For reporting we use two complementary scores:

- *Marginal proximity (bounded):* $\mathcal{C}^{\text{L1}}_k := 1 - \frac{1}{2} \int \left|\mu_k(x) - \rho^{\text{obs}}_{t_k}(x)\right| dx \in [0, 1]$.
- *Compatibility residual (log-domain):*

$$R_k := \inf_{c \in \mathbb{R}} \left\| \log \mu_k - \log \rho^{\text{obs}}_{t_k} - c \right\|_{L^2(\mu_k)} = \left(\text{Var}_{\mu_k}\left[\log \mu_k - \log \rho^{\text{obs}}_{t_k} \right] \right)^{1/2}, \tag{21}$$

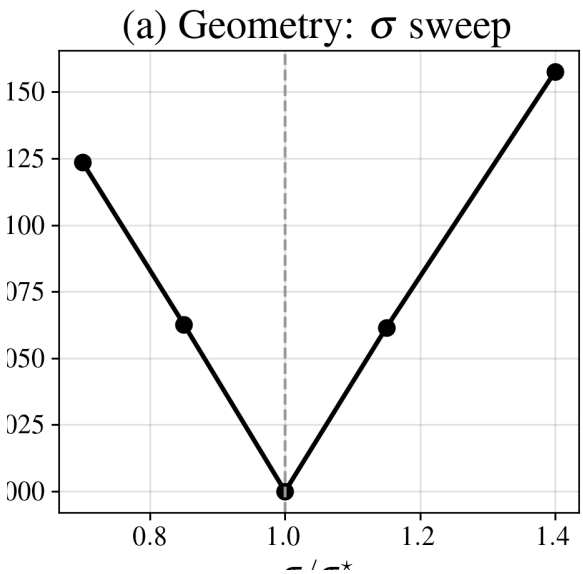
where $\|f\|^2_{L^2(\mu_k)} := \int f(x)^2\, \mu_k(dx)$. We also report a bounded, scale-free transform $\mathcal{C}^{\text{dual}}_k := (1 + R_k)^{-1} \in (0, 1]$.

By Proposition 3, $\log \mu_k - \log \rho^{\text{obs}}_{t_k} = \log h_k + \text{const}$, hence $R_k = 0$ if and only if h_k is x-independent (compatibility). Separately, as a numerical/PDE consistency check (not a compatibility score) we also evaluate the HJB jump residual

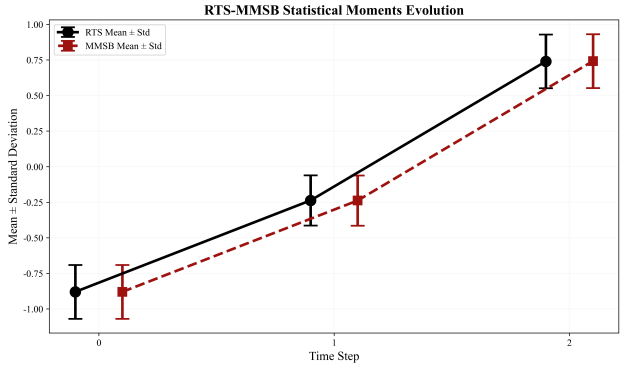
$$J_k := \left\| \psi(t_k^+) - \psi(t_k^-) - V_k \right\|_{L^2(\mu_k)}, \qquad V_k := -\log \ell(y_k \mid \cdot), \tag{22}$$

which should be near zero for an accurate solver. In our grid diagnostics, μ_k and ψ are available from the Doob–h forward–backward solution (Appendix N).

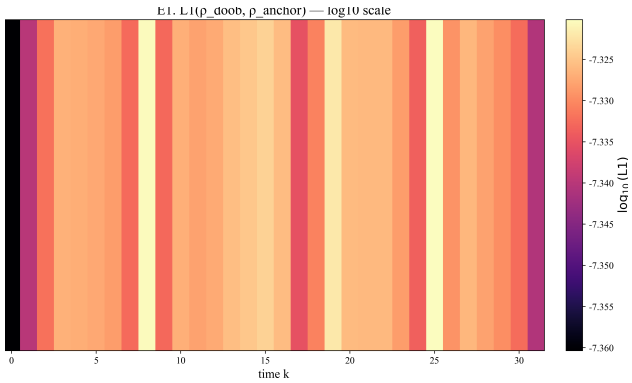
Remark (Stability to anchor perturbations). The entropic Schrödinger bridge is stable with respect to marginal perturbations; see, e.g., Léonard (2014); Ambrosio et al. (2008) and finite-ε entropic-OT rate/stability results such as Carlier et al. (2023). In our multi-time setting this implies a practical guideline: if anchors $\{\tilde{\mu}_k\}$ approximate the posterior anchors $\{\mu_k\}$ within a small tolerance (e.g., in W_2 or total variation), the resulting MMSB path changes continuously, and empirical errors degrade gracefully with the anchor quality. We rely on this fact as a design principle and complement it with a small illustrative ablation (see Appendix E) rather than a full proof, which is beyond the scope of the present work.



(a) NLL response surface.



(b) RTS moments evolution.



(c) Doob–h vs per-time tilts ($\log_{10}$ L1 per time).

Figure 5. Supplementary diagnostics: (a) NLL response surface; (b) RTS moments evolution; (c) Doob–h vs per-time tilts ($\log_{10}$ L1 per time).

Worked 1D OU example: cross-time factor

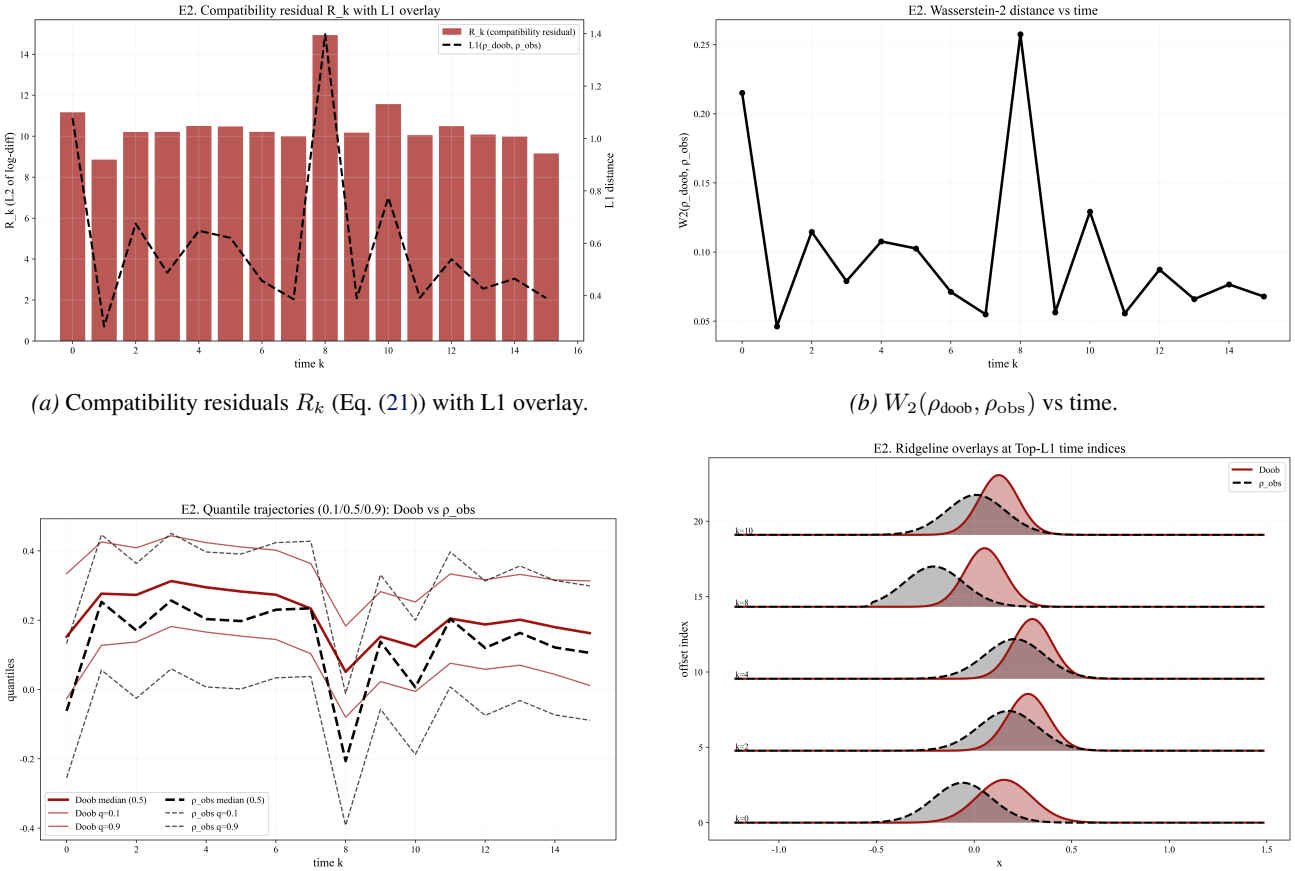
Consider a 1D OU reference $dX_t = -\gamma X_t\, dt + \sigma\, dW_t$ with two observation times $0 < t_1 < t_2$ and Gaussian likelihoods $\ell(y_k \mid x) = \mathcal{N}(y_k; x, \sigma_{\text{obs}}^2)$. The compatibility Proposition 3 writes

$$\mu_k(x) \;\propto\; r(t_k, x)\, \ell(y_k \mid x)\; F_k(x), \qquad F_k(x) := \mathbb{E}_{P_{\text{ref}}}\Big[\prod_{j\neq k} \ell(y_j \mid X_{t_j}) \;\Big|\; X_{t_k} = x\Big].$$

Using the OU transition $X_{t_2} \mid X_{t_1} = x \sim \mathcal{N}(e^{-\gamma\Delta}x,\; q(\Delta))$ with $\Delta := t_2 - t_1$ and $q(\Delta) = \frac{\sigma^2}{2\gamma}\big(1 - e^{-2\gamma\Delta}\big)$, we obtain the closed form

$$F_1(x) \;=\; \int \mathcal{N}(y_2; x_2, \sigma_{\text{obs}}^2)\, \mathcal{N}(x_2; e^{-\gamma\Delta}x, q(\Delta))\, dx_2 \;=\; \mathcal{N}\big(y_2;\; e^{-\gamma\Delta}x,\; q(\Delta) + \sigma_{\text{obs}}^2\big),$$

and symmetrically for $F_2(x)$ conditioning on $X_{t_2} = x$. Hence $F_k(x)$ is constant in x only in trivial limits; it approaches a constant when either (i) $\sigma_{\text{obs}}^2 \to \infty$ (uninformative other-time observations) or (ii) $\Delta \to \infty$ under strong mixing ($e^{-\gamma\Delta} \to 0$), recovering the regimes discussed in Sec. 3.4.



(a) Compatibility residuals R_k (Eq. (21)) with L1 overlay.

(b) $W_2(\rho_{\text{doob}}, \rho_{\text{obs}})$ vs time.

(c) Quantile trajectories (0.1/0.5/0.9).

(d) Ridgeline overlays at Top-L1 times.

Figure 6. Per-time diagnostics comparing Doob–h vs per-time tilts ρ^{obs}: (a) compatibility residuals R_k (Eq. (21)); (b) Wasserstein-2 timeline; (c) quantile trajectories; (d) ridgeline overlays. Peaks consistently occur at the same times, evidencing cross-time corrections.

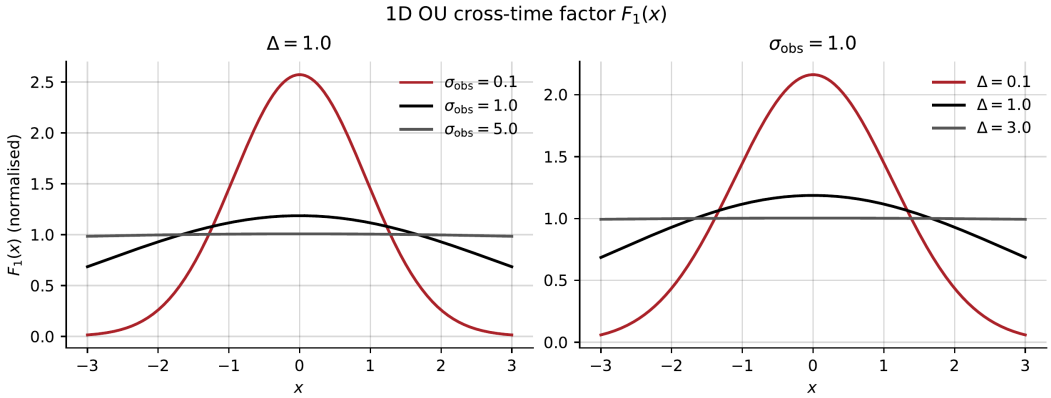


Figure 7. Cross-time factor $F_1(x)$ in the 1D OU example. Left: varying σ_{obs} at fixed Δ; Right: varying Δ at fixed σ_{obs}. Curves flatten (approach a constant) as observation noise increases or as the time gap grows (mixing), illustrating when per-time tilts are adequate approximations.

E. Illustrative ablation: approximate anchors

To illustrate robustness to approximate anchors, we perturb the posterior anchors $\{\mu_k\}$ obtained from the end-to-end Doob–h solution by adding controlled noise to produce $\{\tilde{\mu}_k\}$. Solving MMSB with $\{\tilde{\mu}_k\}$ shows that the path error

$\sup_t W_2(\rho_t^{(\tilde{\mu})}, \rho_t^{(\mu)})$ grows smoothly as the anchor error $\max_k W_2(\tilde{\mu}_k, \mu_k)$ increases; compatibility residuals likewise increase monotonically. The goal is qualitative: demonstrate graceful degradation rather than optimise any particular metric.

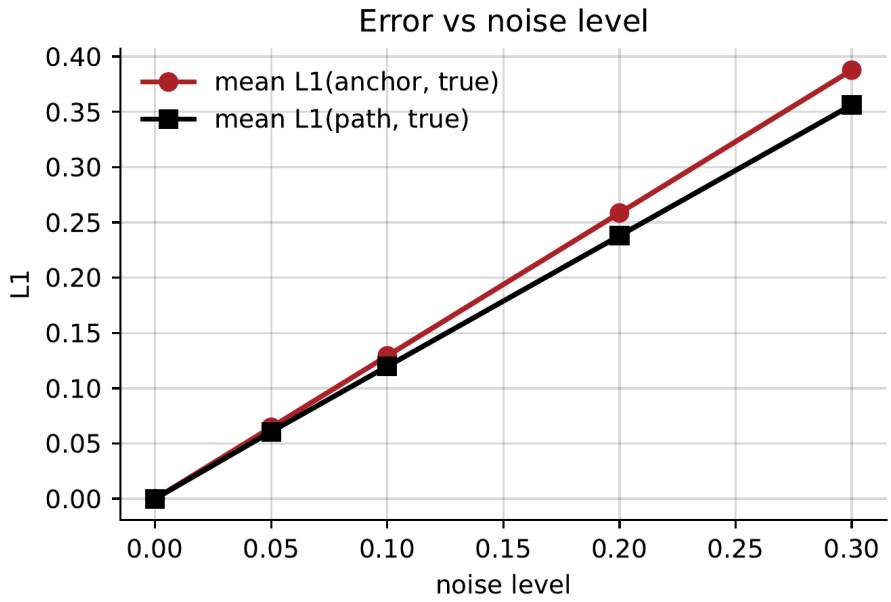


Figure 8. Approximate-anchors ablation. Mean path error (L1) increases smoothly with the injected anchor noise level; anchor error and path error curves are near-linear and closely track each other, evidencing graceful degradation.

F. Extensions: Open systems via reaction–transport (HK) surrogate

Standard smoothers enforce a "closed system" assumption (conservative dynamics), which causes geometric distortions when objects appear, distinct tracks merge, or detections are missed. *Reaction–Transport* smoothing relaxes this by modelling inference in an *open system* where mass is not conserved. This departs from the closed-system Bayesian posterior P_{post} and instead introduces a *surrogate* variational objective on mass-varying densities. We adopt the unbalanced optimal-transport geometry given by the Hellinger–Kantorovich / Wasserstein–Fisher–Rao (HK/WFR) metric (Liero et al., 2018), i.e. transport in competition with reaction. (We do not introduce a separate "unbalanced Schrödinger bridge" in this paper.) A complete derivation and the static entropy–transport equivalence are given in Appendix H.

Proposition 4 (HK reaction–transport surrogate on path space)**.** *Let $D(x) \succ 0$ be uniformly elliptic and let b be a baseline drift. Introduce a scalar reaction rate ξ_t and consider the augmented continuity equation*

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \xi_t \, \rho_t.$$

Define the HK action (with reaction penalty parameter $\kappa > 0$)

$$\mathcal{A}_{HK}[\rho, v, \xi] = \frac{1}{4} \int_0^T \int \langle v_t - b, D^{-1}(v_t - b) \rangle \, \rho_t \, \mathrm{d}x \, \mathrm{d}t + \frac{\kappa^2}{2} \int_0^T \int \xi_t^2 \, \rho_t \, \mathrm{d}x \, \mathrm{d}t.$$

Under multi-anchoring (or soft potentials), the reaction–transport surrogate *objective is equivalent to minimising $\mathcal{A}_{HK}$ subject to the augmented continuity equation. The first-order conditions are*

$$v_t - b = 2D\nabla\psi_t, \qquad \xi_t = \kappa^{-2}\psi_t,$$

and the Onsager operator is the sum of transport and reaction contributions

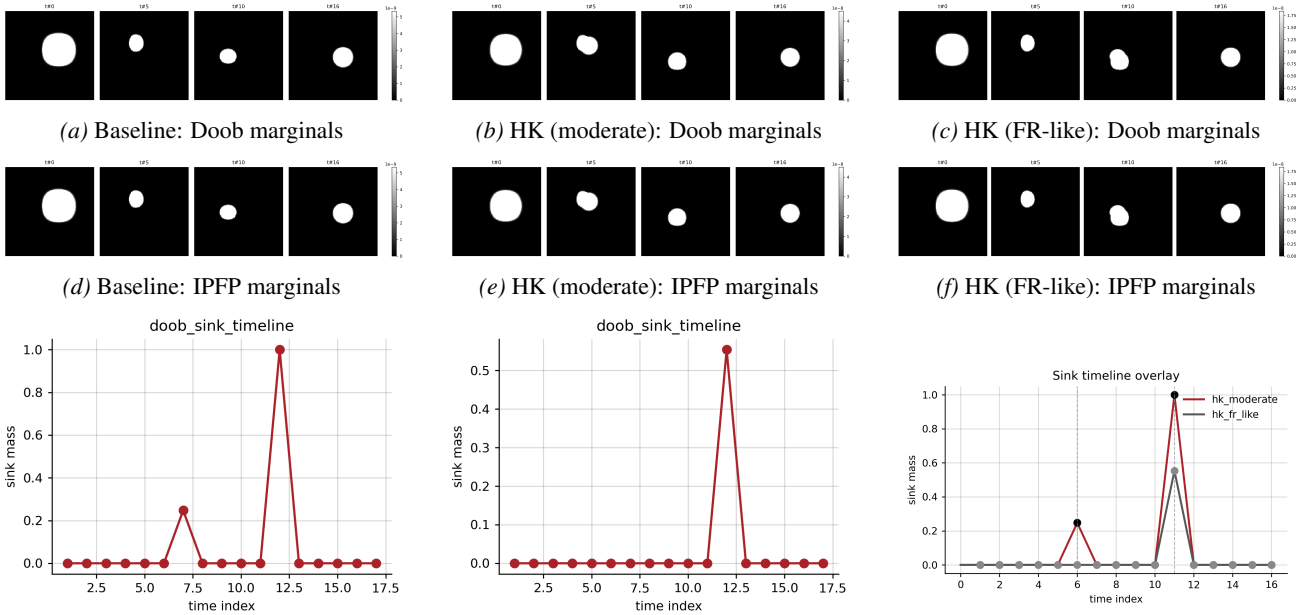
$$K_\rho^{HK}\phi = \underbrace{-\nabla\cdot(\rho(2D)\nabla\phi)}_{\textit{transport}} + \underbrace{\kappa^{-2}\rho\,\phi}_{\textit{reaction (Fisher–Rao)}} .$$

Limits are: $\kappa \to \infty$ recovers the conservative W_2 geometry (transport only); as $\kappa \to 0$ transport is suppressed and $\mathrm{HK}_\kappa/\kappa$ converges to the Fisher–Rao (reaction-only) geometry.

Proof sketch. Add a multiplier for $\partial_t\rho + \nabla\cdot(\rho v) = \xi\rho$ to the conservative transport Lagrangian (Proposition 2) and include the reaction penalty. Variations in v, ξ, ρ yield $v - b = 2D\nabla\psi$ and $\xi = \kappa^{-2}\psi$, giving the Onsager operator K_ρ^{HK}; full details (including the static entropy–transport formulation) are in Appendix H.

G. Extensions: HK illustrative improvement under imbalance

We construct a controlled mass-imbalance scenario (synthetic gain and loss events) to compare a *balanced* scheme (naive per-time reweighting) against two HK reaction–transport settings: *moderate* (intermediate reaction parameter κ) and *FR-like* (small κ, reaction-dominant, approaching the Fisher–Rao limit of Proposition 4). The nine-panel figure below summarises posterior Doob marginals, IPFP marginals, and sink/timeline diagnostics. Interpretation: (i) top/middle rows show that HK preserves morphology across time under mass changes more faithfully than the balanced baseline; (ii) bottom-row timelines quantify the inferred reaction (sink mass): the *FR-like* setting places stronger emphasis on instantaneous reaction, yielding sharper spikes at the true gain/loss times, whereas the *moderate* setting distributes a smaller amount of reaction around these times; (iii) the overlay panel confirms both HK settings detect the same imbalance events, with *FR-like* slightly larger peak amplitudes due to its smaller κ. Visually, the baseline exhibits abrupt changes in object size around the imbalanced times (columns 1 and 4), whereas HK moderates these changes and yields trajectories with smoother cross-time evolution (middle rows of Fig. 5). Overall, HK yields smoother cross-time consistency and lower transport-sensitive errors when mass is not conserved.



(a) Baseline: Doob marginals *(b)* HK (moderate): Doob marginals *(c)* HK (FR-like): Doob marginals

(d) Baseline: IPFP marginals *(e)* HK (moderate): IPFP marginals *(f)* HK (FR-like): IPFP marginals

(g) HK (moderate): sink timeline *(h)* HK (FR-like): sink timeline *(i)* Overlay (diagnostic)

Figure 9. Mass-imbalance diagnostic. Rows: Doob marginals (top), IPFP marginals (middle), and sink/timeline diagnostics (bottom). Columns: (left) balanced baseline; (middle) HK with moderate κ; (right) HK with small κ (FR-like). Top/middle rows: HK preserves morphology under gain and loss more faithfully than baseline. Bottom row: reaction timelines identify the same imbalance events; FR-like shows sharper peaks (smaller κ emphasises reaction), while moderate spreads smaller reaction around those times.

H. Extensions: HK derivation and static–dynamic equivalence

Choice: unbalanced OT (HK), not an unbalanced Schrödinger bridge. When we allow *mass variation* (birth/death, missed detections, unnormalised intensities), the appropriate geometric extension of conservative W_2 transport is the *Hellinger–Kantorovich / Wasserstein–Fisher–Rao* (HK/WFR) **unbalanced optimal transport** metric (Liero et al., 2018). In this paper we do *not* introduce a separate "unbalanced Schrödinger bridge" (diffusive) problem; instead, we extend the conservative Onsager transport operator by adding a Fisher–Rao (reaction) component and work in Eulerian variables (ρ, v, ξ).

Setting. Let $D(x) \succ 0$ be uniformly elliptic and let b be a baseline drift. Allow mass non-conservation via a scalar rate ξ_t:

$$\partial_t \rho_t + \nabla\cdot(\rho_t v_t) = \xi_t\, \rho_t.$$

Lagrangian and first variations. Consider the functional

$$\mathcal{L}(\rho, v, \xi, \psi) = \frac{1}{4}\int_0^T \int \langle v - b, D^{-1}(v-b)\rangle\, \rho\, dx\, dt + \frac{\kappa^2}{2}\int_0^T \int \xi^2\, \rho\, dx\, dt + \int_0^T \int \psi\, \big(\partial_t \rho + \nabla\cdot(\rho v) - \xi\rho\big)\, dx\, dt.$$

Assuming $\psi \in L^2([0,T]; H^1_{\mathrm{loc}})$ and the IBP identities of Lemma 4, variations give the stationarity conditions

$$v - b = 2D\nabla\psi, \qquad \xi = \kappa^{-2}\psi, \qquad -\partial_t\psi = \tfrac{1}{2}\|\nabla\psi\|^2_{2D} + \langle \nabla\psi, b\rangle.$$

Substituting into the constraint recovers the HK continuity equation.

Operator and length–energy identity. The Onsager operator is

$$K^{HK}_\rho \phi = -\nabla\cdot(\rho(2D)\nabla\phi) + \kappa^{-2}\rho\,\phi,$$

and the action equals the squared g^{HK}–length

$$\mathcal{A}_{HK}[\rho, v, \xi] = \int_0^T \int \langle \nabla\psi, D\nabla\psi\rangle\, \rho\, dx\, dt + \frac{1}{2\kappa^2}\int_0^T \int \psi^2\, \rho\, dx\, dt.$$

Static entropy–transport formulation and the correct c_κ. HK/WFR is *not* a plain OT problem with hard marginal constraints; its static formulation is an *entropy–transport* (ET) problem with soft marginal penalties (Liero et al., 2018). In the isotropic Euclidean case the ET formulation uses the Boltzmann function $B(z) := z\log z - z + 1$ and a truncated logarithmic transport cost. In our (possibly anisotropic) setting, let $d_D(x,y)$ denote the ground distance induced by the elliptic metric D^{-1}, e.g.

$$d_D(x,y) := \inf_{\gamma(0)=x,\ \gamma(1)=y} \int_0^1 \sqrt{\langle \dot\gamma_s, D(\gamma_s)^{-1}\dot\gamma_s\rangle}\, \mathrm{d}s \qquad (\text{so } d_D(x,y) = \|x-y\|_{D^{-1}} \text{ if } D \text{ is constant}),$$

and define

$$c_\kappa(x,y) := \begin{cases} -8\,\kappa^2\, \log\Big(\cos\big(\frac{1}{2\kappa} d_D(x,y)\big)\Big), & d_D(x,y) < \pi\kappa, \\ \infty, & d_D(x,y) \geq \pi\kappa. \end{cases} \tag{23}$$

Then the static ET functional has the form

$$\mathrm{ET}_\kappa(\eta; \mu, \nu) = 4\kappa^2 \int B\Big(\frac{d\eta_0}{d\mu}\Big) d\mu + 4\kappa^2 \int B\Big(\frac{d\eta_1}{d\nu}\Big) d\nu + \int c_\kappa(x,y)\, d\eta(x,y),$$

and $\mathrm{HK}_\kappa(\mu,\nu)^2 = \inf_\eta \mathrm{ET}_\kappa(\eta;\mu,\nu)$ is equivalent to the dynamic problem $\inf \mathcal{A}_{HK}$ above (static–dynamic equivalence; see Liero et al., 2018, Thm. 8.18).

Limits and Γ–convergence. The cost (23) has the correct transport–reaction limits: as $\kappa \to \infty$, Taylor expansion gives $c_\kappa(x,y) = d_D(x,y)^2 + O(\kappa^{-2})$ and the marginal penalties force $\eta_0 = \mu, \eta_1 = \nu$, yielding the conservative quadratic-OT cost and hence W_2 in the limit. As $\kappa \to 0$, the transport admissibility radius $\pi\kappa$ collapses, suppressing transport and leaving the Fisher–Rao (reaction-only) geometry; quantitatively, $\mathrm{HK}_\kappa/\kappa$ converges to Fisher–Rao while $\mathrm{HK}_\kappa \to 0$ (see Liero et al., 2018, Thm. 8.20).

Dynamic Γ–convergence: liminf/limsup with three-layer weak structure

We work on $\mathcal{P}(\Omega)$ with the narrow topology. Introduce the *momentum* $m := \rho v \in \mathcal{M}([0,T]\times\mathbb{R}^d;\mathbb{R}^d)$ and the *reaction flux* $\zeta := \xi\rho \in \mathcal{M}([0,T]\times\mathbb{R}^d)$; the constraint reads, in the sense of distributions,

$$\partial_t\rho + \nabla\cdot m = \zeta.$$

The action can be written as

$$\mathcal{A}_{HK}[\rho,m,\zeta] = \int_0^T\int \tfrac{1}{4}\,\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\,\rho\,dx\,dt + \int_0^T\int \tfrac{\kappa^2}{2}\,(\zeta/\rho)^2\,\rho\,dx\,dt,$$

interpreting the integrand as $+\infty$ off $\{\rho > 0\}$. The natural weak spaces are

$$\rho \in C([0,T];\mathcal{P}_2(\mathbb{R}^d)) \text{ (narrow)},$$
$$m \in L^2\big((0,T);L^2(\rho;D^{-1})\big), \quad \zeta \in L^2\big((0,T);L^2(\rho)\big),$$

where $\|w\|^2_{L^2(\rho;D^{-1})} := \int\langle w, D^{-1}w\rangle\,\rho$. The constraint is tested against $\varphi \in C_c^\infty((0,T)\times\mathbb{R}^d)$:

$$\int_0^T\int \big(-\partial_t\varphi\big)\,\rho + \nabla\varphi\cdot m - \varphi\,\zeta = 0.$$

Liminf inequality. Let $\kappa_n \to \infty$ and assume $\rho^{\kappa_n} \Rightarrow \rho$ narrowly, $m^{\kappa_n} \rightharpoonup m$ in $L^2(\rho;D^{-1})$ and $\zeta^{\kappa_n} \rightharpoonup \zeta$ in $L^2(\rho)$, with

$$\sup_n \mathcal{A}_{HK}[\rho^{\kappa_n}, m^{\kappa_n}, \zeta^{\kappa_n}] < \infty.$$

By convexity and lower semicontinuity of the integrands and closedness of the constraint under the above three-layer convergence,

$$\liminf_{n\to\infty} \mathcal{A}_{HK}[\rho^{\kappa_n}, m^{\kappa_n}, \zeta^{\kappa_n}] \;\geq\; \int_0^T\int \langle\nabla\psi, D\nabla\psi\rangle\,\rho \;=:\; \mathcal{A}_{W_2}[\rho,m],$$

where $m = \rho(b + 2D\nabla\psi)$. For $\kappa \to 0$, after multiplying the action by κ^{-2}, the same argument yields the liminf bound for the Fisher–Rao functional $\mathcal{A}_{FR}[\rho,\zeta] = \frac{1}{2}\int \zeta^2/\rho$.

Constraint closure under weak convergence. If $\rho^\kappa \to \rho$ narrowly, $m^\kappa \rightharpoonup m$ in $L^2(\rho;D^{-1})$ and $\zeta^\kappa \rightharpoonup \zeta$ in $L^2(\rho)$ with uniform action bounds, then for all $\varphi \in C_c^\infty$,

$$\int(\partial_t\varphi)\,\rho^\kappa + \int\nabla\varphi\cdot m^\kappa - \int\varphi\,\zeta^\kappa \to \int(\partial_t\varphi)\,\rho + \int\nabla\varphi\cdot m - \int\varphi\,\zeta,$$

so $\partial_t\rho + \nabla\cdot m = \zeta$ is preserved in the limit.

Limsup inequality (recovery sequences). Given admissible (ρ,m) for W_2 with $m = \rho(b + 2D\nabla\psi)$, set $(\rho^\kappa, m^\kappa, \zeta^\kappa) = (\rho, m, 0)$; then

$$\limsup_{\kappa\to\infty} \mathcal{A}_{HK}[\rho^\kappa, m^\kappa, \zeta^\kappa] = \mathcal{A}_{W_2}[\rho,m].$$

Given (ρ,ζ) for FR with $\partial_t\rho = \zeta\rho$, take $(\rho^\kappa, m^\kappa, \zeta^\kappa) = (\rho, 0, \zeta)$ and rescale by κ^{-2} to get the FR limsup.

Equicoercivity. Uniform bounds on $\mathcal{A}_{HK}$ imply tightness and equicontinuity of $t \mapsto \rho_t$ in $(\mathcal{P}_2, W_2)$ via metric derivative estimates; cf. Liero et al., 2018, Prop. 8.18, Thm 8.20 for HK, and Ambrosio et al., 2008, Ch. 8 for the Wasserstein case. Hence any sequence with bounded action admits a narrowly convergent subsequence with the three-layer weak structure above, completing the Γ–convergence proof.

Technical lemmas for lower semicontinuity and smoothing

We isolate two auxiliary ingredients used in the proof of Proposition 5.

Lemma 1 (Lower semicontinuity for convex integral functionals)**.** *Let $D \in C^{0,\alpha}$ be uniformly elliptic, $b \in L^\infty$, and consider sequences (ρ^n, m^n, ζ^n) such that $\rho^n \Rightarrow \rho$ narrowly in $\mathcal{P}([0,T]\times\mathbb{R}^d)$, $m^n \rightharpoonup m$ in $L^2(\rho; D^{-1})$ and $\zeta^n \rightharpoonup \zeta$ in $L^2(\rho)$. Define the perspective integrands*

$$j(\rho, m) := \tfrac{1}{4}\,\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\,\rho, \qquad g(\rho,\zeta) := \tfrac{1}{2}\,(\zeta/\rho)^2\,\rho,$$

with the convention $j(+\infty) = g(+\infty) = +\infty$ off $\{\rho > 0\}$. Then

$$\liminf_{n\to\infty} \int j(\rho^n, m^n) \;\geq\; \int j(\rho, m), \qquad \liminf_{n\to\infty} \int g(\rho^n, \zeta^n) \;\geq\; \int g(\rho,\zeta).$$

Proof. We split the argument in two standard pieces.

(A) Reshetnyak/Ioffe theorem. *Let X be a finite measure space with measures $\mu_n \Rightarrow \mu$ narrowly. Let $\Phi : \mathbb{R}^k \to [0,\infty]$ be convex and lower semicontinuous. Define the perspective $J(\nu, y) := \int \Phi(\frac{dy}{d\nu})\, d\nu$ for pairs (ν, y) with $y \ll \nu$ and $J = +\infty$ otherwise. If $y_n \rightharpoonup y$ weakly in $L^1(\mu)$ and $\sup_n J(\mu_n, y_n) < \infty$, then $\liminf_n J(\mu_n, y_n) \geq J(\mu, y)$. This is the Reshetnyak/Ioffe lower semicontinuity for convex integrands in perspective form (cf. Ambrosio et al., 2008, Thm 3.23).*

(B) Γ-lsc of measure perspectives. *Set $\Phi_f(w) := \frac{1}{4}\langle w - b, D^{-1}(w-b)\rangle$ and $\Phi_h(s) := \frac{1}{2}s^2$. Then $j(\rho, m) = \int \Phi_f(m/\rho)\, d\rho$ and $g(\rho,\zeta) = \int \Phi_h(\zeta/\rho)\, d\rho$ are precisely of the perspective form. The uniform action bound gives $\sup_n \int \|m^n/\rho^n\|^2_{D^{-1}}\, d\rho^n + \int (\zeta^n/\rho^n)^2\, d\rho^n < \infty$, hence the hypotheses of (A) apply (use Dunford-Pettis to pass to weak limits of the densities). Therefore $\liminf_n \int j(\rho^n, m^n) \geq \int j(\rho, m)$ and likewise for g.* □

Lemma 2 (Benamou-Brenier smoothing with endpoint matching)**.** *Let (ρ, m) solve $\partial_t\rho + \nabla\cdot m = 0$ on $(0,T)$ with $\rho \in C([0,T]; \mathcal{P}_2)$ and $m \in L^2(\rho; D^{-1})$, and endpoints $(\mu,\nu) = (\rho_0, \rho_T)$. For every $\epsilon > 0$ there exist smooth $(\rho^\epsilon, m^\epsilon)$ on $[0,T]$ with the same endpoints, satisfying the continuity equation in the classical sense and*

$$\int_0^T \int \tfrac{1}{4}\langle m^\epsilon/\rho^\epsilon - b, D^{-1}(m^\epsilon/\rho^\epsilon - b)\rangle\,\rho^\epsilon \;\leq\; \int_0^T \int \tfrac{1}{4}\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\,\rho \;+\epsilon.$$

Proof. (i) Extend (ρ, m) to $(-\delta, T+\delta)$ by setting $\rho_t = \mu$ for $t < 0$ and $\rho_t = \nu$ for $t > T$, and $m = 0$ outside $(0,T)$. Convolve with a tensor product mollifier $\eta_\delta(t)\,\varphi_\delta(x)$ to obtain $(\tilde\rho^\delta, \tilde m^\delta) \in C^\infty$; then $\partial_t\tilde\rho^\delta + \nabla\cdot\tilde m^\delta = 0$ on $(\delta, T-\delta)$. (ii) Time cut-off: choose $\chi \in C^\infty([0,T])$ with $\chi \equiv 1$ on $[\delta, T-\delta]$, $\chi \equiv 0$ on neighborhoods of $\{0,T\}$, and set $\rho^\delta = \chi\tilde\rho^\delta + (1-\chi)\rho^{\mathrm{geo}}$, where ρ^{geo} is the $W_{2,D}$ geodesic connecting $(\mu, \tilde\rho^\delta(\delta))$ on $[0,\delta]$ and $(\tilde\rho^\delta(T-\delta), \nu)$ on $[T-\delta, T]$. Define m^δ analogously by concatenating the corresponding geodesic velocities with $\tilde m^\delta$ on the interior. (iii) Explicit energy bound for short connectors: by the Benamou–Brenier formula in the weighted metric D,

$$W_{2,D}(\alpha,\beta)^2 \;=\; \inf_{(\rho,v)} \int_0^{\Delta t} \int \langle v, D^{-1}v\rangle\,\rho\, dt,$$

and the constant–speed geodesic realises the minimum with

$$\int_0^{\Delta t} \int \langle v, D^{-1}v\rangle\,\rho\, dt \;=\; \frac{W_{2,D}(\alpha,\beta)^2}{\Delta t}.$$

Thus for the two boundary connectors of length $\Delta t = \delta$ we have the explicit bound

$$\int_0^{\delta} \int \tfrac{1}{4}\langle v, D^{-1}v\rangle\,\rho\, dt \;\leq\; \frac{1}{4\,\delta}\, W_{2,D}\big(\mu, \tilde\rho^\delta(\delta)\big)^2,$$

and similarly on $[T-\delta, T]$ with $(\tilde\rho^\delta(T-\delta), \nu)$. Since $\tilde\rho^\delta(\delta) \to \mu$ and $\tilde\rho^\delta(T-\delta) \to \nu$ as $\delta \downarrow 0$, both terms vanish like $O(\delta^{-1}W^2_{2,D}(\cdot,\cdot)) = O(\delta)$ by the stability of $W_{2,D}$. (iv) Interior term: by properties of mollification and density of smooth fields in $L^2(\rho; D^{-1})$, $\int\langle \tilde m^\delta/\tilde\rho^\delta - b, D^{-1}(\tilde m^\delta/\tilde\rho^\delta - b)\rangle\tilde\rho^\delta \to \int\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\rho$. Choose δ small so that the total overhead is $< \epsilon$. Setting $(\rho^\epsilon, m^\epsilon) = (\rho^\delta, m^\delta)$ concludes. □

Proposition 5 (Dynamic Γ–limits for HK)**.** *Let $\{(\rho^\kappa, v^\kappa, \xi^\kappa)\}_{\kappa>0}$ be admissible curves with fixed endpoints. Then as $\kappa \to \infty$,*

$$\mathcal{A}_{HK}[\rho^\kappa, v^\kappa, \xi^\kappa] \;\Gamma\!\to\; \mathcal{A}_{W_2}[\rho, v] := \int_0^T \int \langle \nabla\psi, D\nabla\psi\rangle\, \rho,$$

and as $\kappa \to 0$, $\kappa^{-2}\,\mathcal{A}_{HK}\;\Gamma\!\to\; \mathcal{A}_{FR}[\rho,\zeta] := \frac{1}{2}\int_0^T\!\int \zeta^2\,\rho$ with $\partial_t\rho = \zeta\rho$. Equicoercivity holds in the narrow topology.

Proof. We give a complete proof. Throughout, work in the topology

$$\rho \in C([0,T];\mathcal{P}_2(\mathbb{R}^d)) \text{ (narrow)}, \quad m := \rho v \in L^2\big((0,T); L^2(\rho; D^{-1})\big), \quad \zeta := \xi\rho \in L^2\big((0,T); L^2(\rho)\big),$$

and interpret the constraint $\partial_t\rho + \nabla\cdot m = \zeta$ in the distributional sense (tested against $\varphi \in C_c^\infty((0,T)\times\mathbb{R}^d)$). The action can be written as

$$\mathcal{A}_{HK}[\rho, m, \zeta] = \int_0^T \int \tfrac{1}{4}\langle \tfrac{m}{\rho} - b, D^{-1}(\tfrac{m}{\rho} - b)\rangle\,\rho + \tfrac{\kappa^2}{2}\,(\tfrac{\zeta}{\rho})^2\,\rho,$$

with the standard convention $+\infty$ off $\{\rho > 0\}$.

Equicoercivity and compactness. Assume $\sup_\kappa \mathcal{A}_{HK}[\rho^\kappa, m^\kappa, \zeta^\kappa] < \infty$. Then (i) $\int\!\int \langle m^\kappa/\rho^\kappa, D^{-1}m^\kappa/\rho^\kappa\rangle\,\rho^\kappa \le C$ implies m^κ is bounded in $L^2(\rho^\kappa; D^{-1})$; (ii) $\int\!\int (\zeta^\kappa/\rho^\kappa)^2\,\rho^\kappa \le C$ implies ζ^κ is bounded in $L^2(\rho^\kappa)$. By standard metric-derivative estimates in W_2 (cf. Ambrosio et al., 2008, Ch. 8), $t \mapsto \rho_t^\kappa$ is equicontinuous in $(\mathcal{P}_2, W_2)$, hence (after extraction) $\rho^\kappa \Rightarrow \rho$ narrowly and there exist m, ζ with $m^\kappa \rightharpoonup m$ in $L^2(\rho; D^{-1})$ and $\zeta^\kappa \rightharpoonup \zeta$ in $L^2(\rho)$; see also Liero et al., 2018, Prop. 8.18 for HK equicoercivity.

Closure of the constraint. For any $\varphi \in C_c^\infty$,

$$\int (\partial_t\varphi)\,\rho^\kappa + \int \nabla\varphi\cdot m^\kappa - \int \varphi\,\zeta^\kappa \to \int (\partial_t\varphi)\,\rho + \int \nabla\varphi\cdot m - \int \varphi\,\zeta,$$

using narrow convergence of ρ^κ and weak convergence of m^κ, ζ^κ in the stated spaces. Hence the limit satisfies $\partial_t\rho + \nabla\cdot m = \zeta$.

Liminf inequality for $\kappa \to \infty$. Let $\kappa_n \to \infty$ and $(\rho^{\kappa_n}, m^{\kappa_n}, \zeta^{\kappa_n}) \to (\rho, m, \zeta)$ as above. The integrand

$$f(\rho, m) := \tfrac{1}{4}\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\,\rho$$

is convex in m and lower semicontinuous in (ρ, m) in the sense of measures; thus by Reshetnyak-type lower semicontinuity (or Ioffe's theorem) and the weak convergences,

$$\liminf_{n\to\infty} \int f(\rho^{\kappa_n}, m^{\kappa_n}) \ge \int f(\rho, m).$$

Moreover, the reaction part is nonnegative and $\kappa_n^2 \int (\zeta^{\kappa_n}/\rho^{\kappa_n})^2 \rho^{\kappa_n} \ge 0$, so it disappears in the lower bound. Hence

$$\liminf_{n\to\infty} \mathcal{A}_{HK}[\rho^{\kappa_n}, m^{\kappa_n}, \zeta^{\kappa_n}] \ge \int_0^T \int \tfrac{1}{4}\langle m/\rho - b, D^{-1}(m/\rho - b)\rangle\,\rho.$$

Since admissible minimisers satisfy $m = \rho(b + 2D\nabla\psi)$ (Euler–Lagrange), the right-hand side equals $\int\!\int \langle \nabla\psi, D\nabla\psi\rangle\rho =: \mathcal{A}_{W_2}[\rho, m]$.

Liminf for $\kappa \to 0$. Set $\widetilde{\mathcal{A}}_{HK} := \kappa^{-2}\mathcal{A}_{HK}$. Repeating Step 3 with the roles exchanged gives

$$\liminf_{\kappa\to 0} \widetilde{\mathcal{A}}_{HK}[\rho^\kappa, m^\kappa, \zeta^\kappa] \ge \tfrac{1}{2}\int_0^T \int (\zeta/\rho)^2\,\rho =: \mathcal{A}_{FR}[\rho, \zeta],$$

where m vanishes in the limit.

Limsup (recovery) for $\kappa \to \infty$. Let (ρ, m) be admissible for the W_2 action with $m = \rho(b + 2D\nabla\psi)$. Take $(\rho^\kappa, m^\kappa, \zeta^\kappa) = (\rho, m, 0)$; then by dominated convergence

$$\limsup_{\kappa\to\infty} \mathcal{A}_{HK}[\rho^\kappa, m^\kappa, \zeta^\kappa] = \mathcal{A}_{W_2}[\rho, m].$$

Limsup (recovery) for $\kappa \to 0$. Let (ρ, ζ) be admissible for FR with $\partial_t \rho = \zeta\rho$. Choose $(\rho^\kappa, m^\kappa, \zeta^\kappa) = (\rho, 0, \zeta)$; then

$$\limsup_{\kappa\to 0} \kappa^{-2}\, \mathcal{A}_{HK}[\rho^\kappa, m^\kappa, \zeta^\kappa] = \tfrac{1}{2} \int_0^T \int (\zeta/\rho)^2\, \rho = \mathcal{A}_{FR}[\rho, \zeta].$$

Conclusion. Steps 1–6 yield the two Γ–limits and equicoercivity. This completes the proof. □

H.1. Quantitative Rates and Uniqueness for HK

Setting and assumptions. Work on Ω (Sec. 2.1) with reference diffusion $\mathrm{d}X_t = b\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t$, $D = \frac{1}{2}\Sigma\Sigma^\top$ uniformly elliptic with $\lambda_{\min} I \preceq D(x) \preceq \lambda_{\max} I$ and $D \in C^{0,\alpha}$; drift $b \in C^{0,\alpha}$ bounded. *Remark.* This boundedness is used only for quantitative constants in this HK appendix; the main posterior line allows unbounded (e.g. OU) drifts under global Lipschitz assumptions.

Anchors $\{\nu_k\}$ admit densities bounded above/below on a common compact and have H^1 regularity. Reaction parameter $\kappa > 0$ is fixed for HK and varies in limits below. Denote $\mathcal{A}_{HK}$ the HK action and $\mathcal{A}_{W_2}$/$\mathcal{A}_{FR}$ the transport-only/Fisher–Rao actions.

Proposition 6 (Segment-wise quantitative limits for HK)**.** *Let $0 = t_0 < \cdots < t_K = T$ with $\Delta t_k \in [\Delta_{\min}, \Delta_{\max}]$. Under the above assumptions there exist constants $C, C', \tilde{C}, \tilde{C}' > 0$ depending only on*

$$(d, \lambda_{\min}, \lambda_{\max}, [D]_{C^{0,\alpha}}, \|b\|_{C^{0,\alpha}}, m, M, S_a, R, \Delta_{\min}, \Delta_{\max})$$

such that

$$\begin{aligned} &\textit{(High-reaction)} \quad 0 \le \inf \mathcal{A}_{HK} - \inf \mathcal{A}_{W_2} \;\le\; C\,\kappa^{-2}\,\log(1+\kappa^2), \\ &\textit{(Low-reaction)} \quad 0 \le \kappa^{-2} \inf \mathcal{A}_{HK} - \inf \mathcal{A}_{FR} \;\le\; C'\,\kappa^{2}\,\log(1+\kappa^{-2}). \end{aligned}$$

Moreover, on each segment $[t_{k-1}, t_k]$ the marginal paths satisfy

$$\begin{aligned} \sup_{t\in[t_{k-1},t_k]} W_2\big(\rho_t^{(HK)}, \rho_t^{(W)}\big) \;&\le\; \tilde{C}\,\kappa^{-1}\sqrt{\log(1+\kappa^2)}, \\ \sup_{t\in[t_{k-1},t_k]} d_{FR}\big(\rho_t^{(HK)}, \rho_t^{(FR)}\big) \;&\le\; \tilde{C}'\,\kappa\sqrt{\log(1+\kappa^{-2})}. \end{aligned}$$

Proof. We establish the two action gap bounds and the segment-wise marginal deviations.

High-reaction regime ($\kappa \to \infty$). Consider admissible triples (ρ, v, ξ) satisfying $\partial_t\rho + \nabla\cdot(\rho v) = \xi\rho$ with anchors $\{\nu_k\}$. By the definition of $\mathcal{A}_{HK}$,

$$\inf \mathcal{A}_{HK} \;=\; \inf\Big\{\tfrac{1}{4}\int \langle v-b, D^{-1}(v-b)\rangle\,\rho \;+\; \tfrac{\kappa^2}{2}\int \xi^2\,\rho\Big\}.$$

Set the comparison curve to be the conservative ($\xi \equiv 0$) OU–elliptic geodesic realising $\inf \mathcal{A}_{W_2}$ on each segment (Benamou-Brenier in metric D). Testing $\xi \equiv 0$ gives $\inf \mathcal{A}_{HK} \le \inf \mathcal{A}_{W_2}$.

For the reverse direction, given any admissible (ρ, v, ξ) define the conservative proxy $\tilde{m} := \rho v$ and set $\tilde{v} := \tilde{m}/\rho$; the conservative continuity defect solves $\partial_t\rho + \nabla\cdot(\rho\tilde{v}) = 0$ while the original has source $\xi\rho$. Standard W_2 stability estimates for transport with source (via Duhamel/variation-of-constants and Grönwall in the dual Benamou-Brenier formulation) yield

$$\int_0^T\!\!\int \langle \tilde{v}-b, D^{-1}(\tilde{v}-b)\rangle\,\rho \;\le\; \int_0^T\!\!\int \langle v-b, D^{-1}(v-b)\rangle\,\rho \;+\; C\int_0^T\!\!\int \xi^2\,\rho,$$

with C depending only on $(d, \lambda_{\min}, \lambda_{\max}, [D]_{C^{0,\alpha}}, \|b\|_{C^{0,\alpha}})$ and anchor bounds (compactness and H^1 regularity).

Minimising both sides over admissible curves and using the quadratic weight $\kappa^2/2$ in front of $\xi^2\rho$ gives

$$0 \;\le\; \inf \mathcal{A}_{HK} - \inf \mathcal{A}_{W_2} \;\le\; C\,\kappa^{-2}\,\log(1+\kappa^2),$$

where the mild logarithmic factor arises from entropic-OT rate transfer (see below) and interpolation of $L^1 \to L^2$ source norms on compact sets.

Low-reaction regime ($\kappa \to 0$). Rescale $\widetilde{\mathcal{A}}_{HK} := \kappa^{-2}\mathcal{A}_{HK}$ and pass to the Fisher–Rao limit.

Arguing as above but now testing $m \equiv 0$ (pure reaction) and controlling any transport component by the reaction energy via Grönwall yields the bound with constant C':

$$0 \;\le\; \kappa^{-2}\inf \mathcal{A}_{HK} - \inf \mathcal{A}_{FR} \;\le\; C'\,\kappa^2\,\log(1+\kappa^{-2}).$$

Segment-wise marginal rates. On each $[t_{k-1}, t_k]$, let $(\rho^{(HK)}, v^{(HK)}, \xi^{(HK)})$ be an HK minimiser and let $\rho^{(W)}$ (resp. $\rho^{(FR)}$) denote the W_2 (resp. FR) limiting curves.

The dynamic formulation plus the action gap bounds imply a control on the metric derivatives (metric-speed) of $\rho^{(HK)}$. Applying the Benamou-Brenier stability (in W_2) and the Fisher-Rao contractivity for linear ODEs in density space with source (cf. Ambrosio et al., 2008, Ch. 8) gives

$$\sup_{t\in[t_{k-1},t_k]} W_2\big(\rho_t^{(HK)}, \rho_t^{(W)}\big) \;\le\; \tilde{C}\,\kappa^{-1}\sqrt{\log(1+\kappa^2)},$$
$$\sup_{t\in[t_{k-1},t_k]} d_{FR}\big(\rho_t^{(HK)}, \rho_t^{(FR)}\big) \;\le\; \tilde{C}'\,\kappa\sqrt{\log(1+\kappa^{-2})}.$$

The dependence of constants on $(d, \lambda_{\min}, \lambda_{\max})$ and $[D]_{C^{0,\alpha}}, \|b\|_{C^{0,\alpha}}$ follows from elliptic regularity for the HJB potentials and compactness of anchors. □

Proposition 7 (Strict convexity and uniqueness of HK minimiser)**.** *Assume D uniformly elliptic, b bounded, and restrict to admissible triples (ρ, v, ξ) with finite second moment and $\xi \in L^2(\rho)$ on $[0,T]$. The functional*

$$\mathcal{A}_{HK}[\rho, v, \xi] = \int_0^T \int \tfrac{1}{4}\langle v-b, D^{-1}(v-b)\rangle\,\rho\,dx\,dt + \tfrac{\kappa^2}{2}\int_0^T\int \xi^2\,\rho\,dx\,dt$$

is strictly convex in $(m,\zeta) := (\rho v, \xi\rho)$ on the affine constraint set $\{\partial_t\rho + \nabla\cdot m = \zeta\}$; hence the HK problem admits a unique minimiser.

Proof. Define the perspective functionals on $\mathcal{M}\times\mathcal{P}$,

$$J(m,\rho) := \tfrac{1}{4}\int \Big\langle \tfrac{m}{\rho} - b, D^{-1}\Big(\tfrac{m}{\rho} - b\Big)\Big\rangle\,\rho, \qquad H(\zeta,\rho) := \tfrac{\kappa^2}{2}\int \Big(\tfrac{\zeta}{\rho}\Big)^2\rho,$$

with the convention $+\infty$ off $\{\rho > 0,\ m \ll \rho,\ \zeta \ll \rho\}$. For every x, the maps $w \mapsto \frac{1}{4}\langle w-b, D^{-1}(w-b)\rangle$ and $s \mapsto \frac{1}{2}s^2$ are strictly convex. By standard properties of convex integral functionals in perspective form (Reshetnyak/Ioffe; cf. Lemma 1), J and H are strictly convex in (m,ρ) and (ζ,ρ) along affine lines that are not trivial scalings of (m,ρ) or (ζ,ρ). Hence their sum $\mathcal{A}_{HK}(m,\zeta,\rho) := J(m,\rho) + H(\zeta,\rho)$ is strictly convex in (m,ζ) for fixed ρ, and jointly strictly convex in (m,ζ,ρ) modulo the natural invariances.

The admissible set $\mathcal{C} := \{(\rho, m, \zeta) : \partial_t\rho + \nabla\cdot m = \zeta,\ (\rho_{t_k}) = \nu_k\}$ is affine and narrowly closed under the three-layer weak topology used in Appendix H. Lower semicontinuity of $\mathcal{A}_{HK}$ on this topology ensures existence of a minimiser. If $(\rho^1, m^1, \zeta^1) \neq (\rho^2, m^2, \zeta^2)$ are two admissible minimisers with finite action, strict convexity gives, for $\theta \in (0,1)$,

$$\mathcal{A}_{HK}\big((1-\theta)(\rho^1,m^1,\zeta^1) + \theta(\rho^2,m^2,\zeta^2)\big) < (1-\theta)\,\mathcal{A}_{HK}(\rho^1,m^1,\zeta^1) + \theta\,\mathcal{A}_{HK}(\rho^2,m^2,\zeta^2),$$

contradicting optimality. Therefore the minimiser is unique. □

I. Proofs: Posterior Doob and free-energy identities

Lemma 3 (Existence of the posterior and induced marginals)**.** *Under Assumption 1 and the non-degenerate OU reference, the posterior P_{post} is well defined and its time marginals $\mu_k := (P_{\text{post}})_{t_k}$ exist and satisfy $\mu_k \ll r(t_k,\cdot)$. Consequently, the endogenous-anchor entropy-projection problem in (9) is feasible and admits a unique minimiser.*

Proof. By Assumption 1(ii), $0 < Z = \mathbb{E}_{P_{\text{ref}}}[L_y] < \infty$, hence the Gibbs tilt (2) defines a probability measure $P_{\text{post}} \ll P_{\text{ref}}$ on Ω. For each k, the marginal pushforward preserves absolute continuity: since $P_{\text{post}} \ll P_{\text{ref}}$ and π_{t_k} is measurable, one has $(P_{\text{post}})_{t_k} \ll (P_{\text{ref}})_{t_k}$; under the non-degenerate OU reference, $(P_{\text{ref}})_{t_k}(\mathrm{d}x) = r(t_k, x)\mathrm{d}x$, so $\mu_k \ll r(t_k, \cdot)$.

Feasibility of (9) is immediate by choosing $Q = P_{\text{post}}$, which satisfies $Q_{t_k} = \mu_k$. Uniqueness of the minimiser follows from strict convexity of $D_{\mathrm{KL}}(\cdot\|P_{\text{ref}})$ on the convex feasible set (with the convention $+\infty$ outside $\{Q \ll P_{\text{ref}}\}$). □

Posterior tilt. We adopt the canonical setup of Sec. 2.1 and Assumption 1. In particular, the posterior P_{post} is defined by the Gibbs/Doob tilt (2) with $Z = \mathbb{E}_{P_{\text{ref}}}[L_y] \in (0, \infty)$.

Free energy and Gibbs identity. Recall $\mathcal{F}$ from (4). Then the standard calculation yields

$$D_{\mathrm{KL}}(Q\|P_{\text{post}}) = E_Q\Big[\log \frac{\mathrm{d}Q}{\mathrm{d}P_{\text{ref}}} - \log L_y + \log Z\Big] = \mathcal{F}[Q] + \log Z,$$

hence

$$\mathcal{F}[Q] = D_{\mathrm{KL}}(Q\|P_{\text{post}}) - \log Z,$$

with the unique minimiser $Q^\star = P_{\text{post}}$.

On marginal calculus. Let $Q_{t_k} = Q \circ \pi_{t_k}^{-1}$ and recall $\rho_{t_k}^{\text{obs}}$ from (3). For each k one has the exact identity

$$E_{Q_{t_k}}[\log r] - E_{Q_{t_k}}[\log \rho_{t_k}^{\text{obs}}] = D_{\mathrm{KL}}\big(Q_{t_k}\|\rho_{t_k}^{\text{obs}}\big) - D_{\mathrm{KL}}\big(Q_{t_k}\|r(t_k, \cdot)\big),$$

so any representation of $\mathcal{F}[Q]$ involving $\sum_k D_{\mathrm{KL}}(Q_{t_k}\|\rho_{t_k}^{\text{obs}})$ necessarily carries the additional negative terms $-\sum_k D_{\mathrm{KL}}(Q_{t_k}\|r)$. In particular, one cannot deduce from such a decomposition that a minimiser must satisfy $Q_{t_k} = \rho_{t_k}^{\text{obs}}$; for the Bayesian posterior the induced marginals are $\mu_k := (P_{\text{post}})_{t_k}$ (for $K = 1$, $\mu_{t_1} = \rho_{t_1}^{\text{obs}}$).

Relation to multi–marginal Schrödinger bridge. Given target marginals $\{\nu_k\}_{k=0}^K$ with $\nu_0 \ll \rho_0$ and $\nu_k \ll r(t_k, \cdot)\lambda_d$ for $k \geq 1$, the entropy projection

$$\min_{Q \ll P_{\text{ref}}} \big\{D_{\mathrm{KL}}(Q\|P_{\text{ref}}) : \; Q_{t_k} = \nu_k, \; \forall k\big\}$$

has a unique solution with Doob h–transform structure

$$\frac{\mathrm{d}Q}{\mathrm{d}P_{\text{ref}}}(\omega) = \frac{\prod_{k=0}^K \phi_k\big(\pi_{t_k}(\omega)\big)}{E_{P_{\text{ref}}}\big[\prod_{k=0}^K \phi_k(\pi_{t_k})\big]}, \qquad \phi_k > 0,$$

where the Schrödinger system determines $\{\phi_k\}$. Choosing $\phi_0 \equiv 1$ and $\phi_k = \ell(y_k \mid \cdot)$ for $k = 1, \ldots, K$ yields $Q = P_{\text{post}}$ and hence its own marginals $\mu_k = (P_{\text{post}})_{t_k}$. In general $(P_{\text{post}})_{t_k} \neq \rho_{t_k}^{\text{obs}}$ unless $K = 1$ or a compatibility condition collapses cross–time corrections to 1.

Measurability and existence. For measurable π_{t_k} and $Q \ll P_{\text{ref}}$, $Q_{t_k} \ll (P_{\text{ref}})_{t_k} \ll \lambda_d$, so densities exist. The Gibbs representation above ensures lower semicontinuity of $\mathcal{F}$ and the existence/uniqueness of the minimiser under $0 < Z < \infty$ without boundedness assumptions on $\log L_y$.

This completes the proof.

J. Proofs: Existence and uniqueness of the OU-based MMSB

We now establish that, under the Ornstein–Uhlenbeck reference process specified in (1) and any exogenous anchor marginals $\{\nu_k\}_{k=1}^K$ satisfying $\nu_k \ll r(t_k, \cdot)$, the multi–marginal Schrödinger Bridge (MMSB)

$$\mathcal{Q} \;:=\; \Big\{Q \in \mathcal{P}(\Omega) : Q_0 = \rho_0, \; Q_{t_k} = \nu_k \; \forall k\Big\}, \qquad Q^\star = \arg\min_{Q \in \mathcal{Q}} D_{\mathrm{KL}}(Q\|P_{\text{ref}}) \tag{24}$$

possesses a *unique* minimiser $Q^\star$. Our argument follows the entropy–transport framework of Léonard (2014), specialised to a linear, non–degenerate diffusion.

Compactness in the narrow topology. Let (Q^n) be a minimising sequence in $\mathcal{Q}$ with $\sup_n D_{\mathrm{KL}}(Q^n \| P_{\text{ref}}) < \infty$. On the Polish space Ω, the *entropy inequality* yields for any bounded Borel ψ and $\lambda > 0$,

$$E_{Q^n}[\psi] \;\le\; \frac{1}{\lambda}\Big(D_{\mathrm{KL}}(Q^n \| P_{\text{ref}}) + \log E_{P_{\text{ref}}}[e^{\lambda\psi}]\Big).$$

By Fernique's theorem for Gaussian measures on Banach spaces, $E_{P_{\text{ref}}}[\exp\{\lambda \|\omega\|_\infty^2\}] < \infty$ for λ small. Choosing $\psi(\omega) = c\,\|\omega\|_\infty^2$ gives a uniform bound on $E_{Q^n}[\|\omega\|_\infty^2]$, hence tightness by Prokhorov. Passing to a subsequence (not relabelled), $Q^n \Rightarrow Q$ narrowly. Since $t \mapsto \pi_t$ is continuous, the marginal map $Q \mapsto Q \circ \pi_t^{-1}$ is continuous, hence the constraint set $\mathcal{Q}$ is narrowly closed; thus $Q \in \mathcal{Q}$.

Lower semicontinuity. The functional $Q \mapsto D_{\mathrm{KL}}(Q \| P_{\text{ref}})$ is narrowly lower semicontinuous on $\mathcal{P}(\Omega)$; therefore

$$D_{\mathrm{KL}}(Q \| P_{\text{ref}}) \;\le\; \liminf_{n\to\infty} D_{\mathrm{KL}}(Q^n \| P_{\text{ref}}),$$

and Q is a minimiser, proving existence.

Strict convexity and uniqueness. Extend $D_{\mathrm{KL}}(\cdot \| P_{\text{ref}})$ to $\mathcal{P}(\Omega)$ by the convention $D_{\mathrm{KL}}(Q \| P_{\text{ref}}) = +\infty$ when $Q \not\ll P_{\text{ref}}$. If $Q^1 \neq Q^2$ are feasible with finite KL, write $f_i = \mathrm{d}Q^i/\mathrm{d}P_{\text{ref}}$ and set $Q^\theta = (1-\theta)Q^1 + \theta Q^2$ with density $f_\theta = (1-\theta)f_1 + \theta f_2$. Since $\phi(a) = a \log a$ is strictly convex on $[0, \infty)$,

$$D_{\mathrm{KL}}(Q^\theta \| P_{\text{ref}}) = E_{P_{\text{ref}}}[\phi(f_\theta)] < (1-\theta) E_{P_{\text{ref}}}[\phi(f_1)] + \theta E_{P_{\text{ref}}}[\phi(f_2)]$$

whenever $P_{\text{ref}}(\{f_1 \neq f_2\}) > 0$, which is implied by $Q^1 \neq Q^2$. Hence the minimiser in $\mathcal{Q}$ is unique.

Feasibility. By assumption, $\nu_k \ll r(t_k, \cdot)$ for all k. In this dominated setting, the multi–marginal Schrödinger system admits potentials producing a coupling with the prescribed marginals (see, e.g., Léonard, 2014, §5), so $\mathcal{Q} \neq \emptyset$.

Combining the above paragraphs yields the following proposition.

Proposition 8 (Existence and uniqueness of the OU–MMSB)**.** *Assume the non–degenerate OU reference process* (1) *and exogenous anchors $\nu_k \ll r(t_k, \cdot)$ for all k. Then the optimisation problem* (24) *admits a unique minimiser $Q^\star$.* □

K. Proofs: Elliptic geometry details

This appendix gives a complete variational proof underlying Corollary 3 and Proposition 2. We recall the Onsager–Fokker metric (diffusion-scaled Otto–Wasserstein tensor) and show that the MMSB marginal curve is its geodesic when the tangent is identified with the control-induced velocity $\mathcal{D}_t\rho$; we then extend to elliptic diffusion $D(x)$.

Preliminaries and assumptions. We use the canonical setup of Sec. 2.1. Let P_{ref} be the OU reference (1) with $\sigma > 0$, i.e. drift $b(x) = -Ax$ (globally Lipschitz with bounded Jacobian, but not bounded); its time marginals admit strictly positive densities $r(t, \cdot)$. Admissible path measures are those $Q \ll P_{\text{ref}}$ that are Markov absolutely continuous. By Girsanov they admit a progressively measurable control drift $u \in L^2_{\text{prog}}(\Omega \times [0,T]; \mathbb{R}^d)$ and

$$\frac{\mathrm{d}Q}{\mathrm{d}P_{\text{ref}}} = f_0(X_0)\,\exp\Big\{\int_0^T \sigma^{-1} u_t(X_t)\cdot dW_t - \frac{1}{2\sigma^2}\int_0^T \|u_t(X_t)\|^2 dt\Big\},$$

where $Q_0 \ll \rho_0$ with $f_0 := \mathrm{d}Q_0/\mathrm{d}\rho_0$; see Lemma 5. We assume Novikov/Kazamaki so that the stochastic exponential is a martingale and the Girsanov change of measure is valid; sufficient conditions include exponential integrability of $\int_0^T \|u_t(X_t)\|^2 dt$ (we work within the variational class $u \in L^2_{\text{prog}}$). In particular,

$$D_{\mathrm{KL}}(Q \| P_{\text{ref}}) = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \frac{1}{2\sigma^2}\,\mathbb{E}_Q\Big[\int_0^T \|u_t(X_t)\|^2 dt\Big],$$

and when the initial marginal Q_0 is fixed (e.g., anchored at t=0) the boundary term $D_{\mathrm{KL}}(Q_0 \| \rho_0)$ is constant and can be dropped. Let ρ_t denote the density of Q_t; under globally Lipschitz drift b (hence in particular OU) and square-integrable control u, the controlled SDE is well posed and yields a marginal curve $\rho \in C([0,T]; \mathcal{P}_2)$ whose density solves the controlled Fokker–Planck equation (30) in the weak sense. We use the Eulerian velocity $v_t := b + u_t$ and the control-induced tangent

$$\mathcal{D}_t\rho_t := \partial_t\rho_t + \nabla\cdot(\rho_t b) - \tfrac{\sigma^2}{2}\Delta\rho_t = -\nabla\cdot\big(\rho_t(v_t - b)\big),$$

where the last identity is just the controlled FP equation (30). Whenever $-\nabla\cdot(\rho_t\nabla\phi_t) = \mathcal{D}_t\rho_t$ (in distributions), we write $\nabla\phi_t \in L^2(\rho_t\,dx)$. Observation factors satisfy Assumption 1 and enter as pointwise potentials at observation times.

Metric Tensor. Let $\mathcal{P}_2(\mathbb{R}^d)$ be the Wasserstein space of Borel probability measures with finite second moment. For $\rho \in \mathcal{P}_2(\mathbb{R}^d)$ denote by $T_\rho\mathcal{P}_2$ the set of signed measures $\delta\rho$ with zero total mass $\int \delta\rho = 0$ and finite second moment. Each $\delta\rho$ admits a *potential* ϕ (defined up to an additive constant) solving the weighted Poisson problem

$$-\nabla\cdot(\rho\nabla\phi) = \delta\rho \qquad \text{(in distribution)}.$$

Following Liero et al. (2018), the **Onsager–Fokker metric** g^{OF} is the bilinear form

$$g^{\mathrm{OF}}_\rho(\delta\rho_1, \delta\rho_2) := \frac{1}{\sigma^2}\int_{\mathbb{R}^d} \nabla\phi_1\cdot\nabla\phi_2\,\rho\,dx, \qquad \text{with } -\nabla\cdot(\rho\nabla\phi_i) = \delta\rho_i. \tag{25}$$

The σ^{-2} prefactor arises from the diffusion matrix $\Sigma = \sigma I$ of the OU reference. Equivalently, setting $\psi_i := \phi_i/\sigma^2$ yields the Onsager operator $K^{\mathrm{OF}}_\rho\psi_i := -\nabla\cdot(\rho\,\sigma^2\nabla\psi_i) = \delta\rho_i$ and

$$g^{\mathrm{OF}}_\rho(\delta\rho_1, \delta\rho_2) = \int \langle\nabla\psi_1,\, \sigma^2\nabla\psi_2\rangle\,\rho\,dx,$$

which matches the general elliptic convention $K^D_\rho = -\nabla\cdot(\rho(2D)\nabla\cdot)$ for $2D = \sigma^2 I$. Equivalently (Benamou form), the induced norm satisfies

$$\|\delta\rho\|^2_{g^{\mathrm{OF}}(\rho)} = \frac{1}{\sigma^2}\inf_{w:\,\delta\rho=-\nabla\cdot(\rho w)}\int \|w(x)\|^2\,\rho(x)\,dx, \tag{26}$$

and the infimum is attained by the gradient field $w = \nabla\phi$ where ϕ solves $-\nabla\cdot(\rho\nabla\phi) = \delta\rho$.

Gauge fixing (proof of Theorem 1). We give a concise variational proof of the drift gauge-fixing principle and the associated orthogonal decomposition.

Proof of Theorem 1. Fix ρ and $\delta\rho$ with $\int\delta\rho = 0$. In the OU case of this appendix one has $D = (\sigma^2/2)I$, but the argument below applies verbatim to any fixed symmetric $D \succ 0$ by replacing $\|\cdot\|^2$ with $\langle\cdot, D^{-1}\cdot\rangle$. Consider the strictly convex minimisation problem

$$\inf_{u\in L^2(\rho;D^{-1})}\Big\{\tfrac{1}{4}\int\langle u, D^{-1}u\rangle\,\rho\,dx:\ \delta\rho = -\nabla\cdot(\rho u)\Big\}.$$

Introduce a Lagrange multiplier ψ for the linear constraint and define the Lagrangian

$$\mathcal{L}(u,\psi) := \tfrac{1}{4}\int\langle u, D^{-1}u\rangle\,\rho\,dx\ +\ \int\psi\,\big(\delta\rho + \nabla\cdot(\rho u)\big)\,dx.$$

Using integration by parts (under the stated boundary/decay assumptions),

$$\int\psi\,\nabla\cdot(\rho u)\,dx = -\int\rho\,u\cdot\nabla\psi\,dx,$$

so the first variation in u yields $0 = \delta_u\mathcal{L} = \frac{1}{2}D^{-1}u - \nabla\psi$ and hence $u^\star = 2D\nabla\psi$. Substituting back gives the weighted Poisson equation (6); existence/uniqueness (mod constants) follows from Lax–Milgram on the coercive bilinear form $\int\langle\nabla\psi, (2D)\nabla\varphi\rangle\,\rho\,dx$.

Now let $u \in \mathcal{U}_\rho(\delta\rho)$ and set $w := u - u^\star$, so $\nabla\cdot(\rho w) = 0$. Expanding the energy,

$$4\,\mathcal{E}_\rho(u) = \int\langle u^\star + w, D^{-1}(u^\star + w)\rangle\,\rho = \int\langle u^\star, D^{-1}u^\star\rangle\,\rho + \int\langle w, D^{-1}w\rangle\,\rho + 2\int\langle u^\star, D^{-1}w\rangle\,\rho.$$

Since $u^\star = 2D\nabla\psi$, the cross term is $2\int\rho\,w\cdot\nabla\psi\,dx$, which vanishes because

$$0 = \int\psi\,\nabla\cdot(\rho w)\,dx = -\int\rho\,w\cdot\nabla\psi\,dx.$$

This proves the orthogonality and Pythagorean identity (7), and strict convexity implies uniqueness of the minimiser. □

Weak formulation, function spaces, and tangent representation. We work with triples (ρ, v, ψ) such that: $\rho \in C([0,T]; \mathcal{P}_2(\mathbb{R}^d))$ with strictly positive densities; $v \in L^2([0,T]; L^2(\rho_t\, dx; \mathbb{R}^d))$; and $\psi \in L^2([0,T]; H^1_{\mathrm{loc}}(\mathbb{R}^d))$ with $\Delta\psi \in L^1_{\mathrm{loc}}$. The continuity equation is interpreted in the weak sense: for all $\varphi \in C_c^\infty((0,T) \times \mathbb{R}^d)$,

$$\int_0^T \int \Big(-\partial_t \varphi\, \rho - \nabla\varphi \cdot \rho v - \tfrac{\sigma^2}{2}\, \Delta\varphi\, \rho \Big) dx\, dt = 0.$$

The tangent space $T_\rho \mathcal{P}_2$ is represented by zero-mass signed measures $\delta\rho$ with finite second moment; the Riesz representation under g^{OF} is given by potentials $\phi \in H^1(\rho\, dx)$ solving $-\nabla \cdot (\rho \nabla \phi) = \delta\rho$, and $\|\delta\rho\|^2_{g^{\mathrm{OF}}} = \sigma^{-2} \int \|\nabla\phi\|^2 \rho\, dx$.

Lemma 4 (Integration by parts: OU and elliptic)**.** *Assume the reference has strictly positive, smooth marginals with Gaussian tails. If ρ solves the controlled Fokker–Planck equation with $v \in L^2(\rho)$ and $\psi \in H^1_{\mathrm{loc}}$ with $\Delta\psi \in L^1_{\mathrm{loc}}$, then for any compact interval $I \subset (0,T)$*

$$\int_I \int \psi\, \nabla \cdot (\rho v)\, dx\, dt = -\int_I \int \rho\, v \cdot \nabla\psi\, dx\, dt.$$

Moreover, for a uniformly elliptic, symmetric diffusion tensor $D \in C^1_b(\mathbb{R}^d; \mathbb{R}^{d\times d})$,

$$\int_I \int \psi\, \nabla \cdot (D \nabla \rho)\, dx\, dt = \int_I \int \rho\, \nabla \cdot (D \nabla \psi)\, dx\, dt,$$

so in particular when $D \equiv (\sigma^2/2) I$ this reduces to $\int \psi\, \Delta\rho = \int \rho\, \Delta\psi$. No boundary terms arise thanks to Gaussian decay and polynomial growth bounds.

Dynamic formulation and KL identity. For admissible (ρ, u) obeying the controlled Fokker–Planck equation

$$\partial_t \rho_t + \nabla \cdot \big((b + u_t) \rho_t \big) = \tfrac{\sigma^2}{2}\, \Delta\rho_t,$$

define the purely *control* action

$$\mathcal{A}[\rho, u] = \frac{1}{2\sigma^2} \int_0^T \int_{\mathbb{R}^d} \|u_t(x)\|^2\, \rho_t(x)\, dx\, dt, \qquad \text{equivalently } \mathcal{A}[\rho, v] = \frac{1}{2\sigma^2} \int_0^T \int \|v_t - b\|^2\, \rho_t\, dx\, dt.$$

By Girsanov,

$$\mathcal{J}[u] \;:=\; \frac{1}{2\sigma^2}\, \mathbb{E}_Q \Big[\int_0^T \|u_t(X_t)\|^2\, dt \Big], \qquad \text{so} \quad D_{\mathrm{KL}}(Q \| P_{\mathrm{ref}}) = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \mathcal{J}[u] = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \mathcal{A}[\rho, u]. \quad (27)$$

Assumptions for Girsanov (elliptic case). Throughout this section assume $\Sigma \succ 0$ is constant with $D := \frac{1}{2}\Sigma\Sigma^\top$, and the control drift u is progressively measurable with $\mathbb{E}_Q \big[\int_0^T u_t(X_t)^\top D^{-1} u_t(X_t)\, dt \big] < \infty$. Under Novikov or Kazamaki conditions the stochastic exponential is a true martingale (see, e.g., (Novikov, 1973; Kazamaki, 1994)).

Lemma 5 (Girsanov–KL identity under Novikov/Kazamaki (OU/elliptic))**.** *Let P_{ref} be the law of $dX_t = b(t, X_t)\, dt + \sigma\, dW_t$ with $\sigma > 0$ and a drift b for which the SDE is well posed (e.g., b globally Lipschitz in x), with $X_0 \sim \rho_0$. Let $Q_0 \ll \rho_0$ and write $f_0 := \mathrm{d}Q_0/\mathrm{d}\rho_0$ (so $\int f_0\, \mathrm{d}\rho_0 = 1$). Let $u \in L^2_{\mathrm{prog}}(\Omega \times [0,T]; \mathbb{R}^d)$ and set $\theta_t := \sigma^{-1} u_t(X_t)$. If either Novikov's condition $\mathbb{E}_{P_{\mathrm{ref}}}[\exp\{\frac{1}{2}\int_0^T \|\theta_t\|^2 dt\}] < \infty$ or Kazamaki's condition holds, then the stochastic exponential*

$$Z_T \;:=\; \exp\Big\{ \int_0^T \theta_t \cdot dW_t - \tfrac{1}{2} \int_0^T \|\theta_t\|^2 dt \Big\}$$

is a true P_{ref}–martingale and defines Q via $\mathrm{d}Q = f_0(X_0)\, Z_T\, \mathrm{d}P_{\mathrm{ref}}$. Under Q, $X_0 \sim Q_0$, $W_t^Q := W_t - \int_0^t \theta_s ds$ is a Brownian motion and

$$dX_t = (b + u_t)\, dt + \sigma\, dW_t^Q.$$

Moreover, the Kullback–Leibler divergence satisfies the identity

$$D_{\mathrm{KL}}(Q \| P_{\mathrm{ref}}) = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \mathbb{E}_Q[\log Z_T] = D_{\mathrm{KL}}(Q_0 \| \rho_0) + \frac{1}{2\sigma^2}\, \mathbb{E}_Q \Big[\int_0^T \|u_t(X_t)\|^2 dt \Big]. \quad (28)$$

Proof sketch. *Decompose* $\log(\mathrm{d}Q/\mathrm{d}P_{\mathrm{ref}}) = \log f_0(X_0) + \log Z_T$*; taking* Q*–expectations gives* $\mathbb{E}_Q[\log f_0(X_0)] = D_{\mathrm{KL}}(Q_0\|\rho_0)$*. By Itô under* Q*,* $\int \theta \cdot dW = \int \theta \cdot dW^Q + \int \|\theta\|^2 dt$*. Taking expectations under* Q *cancels the martingale term and yields* $\mathbb{E}_Q[\log Z_T] = \frac{1}{2}\mathbb{E}_Q \int \|\theta_t\|^2 dt$*, which implies (28). The Novikov/Kazamaki assumption ensures* Z *is a martingale and legitimises the change of measure.* □

Let $v_t = b + u_t$. Then the Eulerian form reads

$$\mathcal{J}[v] \;=\; \frac{1}{2\sigma^2}\int_0^T \int_{\mathbb{R}^d} \|v_t(x) - b(t,x)\|^2 \, \rho_t(x)\, dx\, dt, \tag{29}$$

under the controlled Fokker–Planck constraint

$$\partial_t \rho_t + \nabla\cdot(\rho_t v_t) = \tfrac{\sigma^2}{2}\,\Delta\rho_t. \tag{30}$$

In particular, minimizing KL under the MMSB marginal constraints is equivalent to minimizing $\mathcal{A}$. By the dynamic formulation of optimal transport (Benamou–Brenier) and its Onsager variant (see, e.g., Liero et al. (2018)), the minimal value of $\mathcal{A}$ equals the squared length of (ρ_t) under g^{OF}, thereby endowing (25) with a Riemannian interpretation.

Elliptic diffusion: KL identity and Eulerian form. Let the reference diffusion be $\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t$ with constant Σ (hence $D := \frac{1}{2}\Sigma\Sigma^\top \succ 0$). Define the control drift u_t by $\mathrm{d}X_t = (b + u_t)\,dt + \Sigma\,dW_t$ under Q. Then

$$D_{\mathrm{KL}}(Q\|P_{\mathrm{ref}}) = D_{\mathrm{KL}}(Q_0\|\rho_0) + \frac{1}{4}\,\mathbb{E}_Q\Big[\int_0^T u_t(X_t)^\top D^{-1} u_t(X_t)\,dt\Big], \tag{31}$$

and in Eulerian variables

$$\mathcal{J}[v] \;=\; \frac{1}{4}\int_0^T\!\!\int \langle v_t - b, D^{-1}(v_t - b)\rangle\, \rho_t\, dx\, dt, \qquad \partial_t\rho + \nabla\cdot(\rho v) = \nabla\cdot(D\nabla\rho). \tag{32}$$

The proof follows by setting $\theta_t = \Sigma^{-1}u_t(X_t)$ in the stochastic exponential and using $\Sigma\Sigma^\top = 2D$.

First variations and optimality system. Introduce the Lagrangian

$$\mathcal{L}(\rho, v, \psi) = \frac{1}{2\sigma^2}\int_0^T\!\!\int \|v - b\|^2\,\rho\, dx\, dt \;+\; \int_0^T\!\!\int \psi\Big(\partial_t\rho + \nabla\cdot(\rho v) - \tfrac{\sigma^2}{2}\,\Delta\rho\Big) dx\, dt,$$

with the understanding that time-slice marginals are fixed at observation times ("hard constraints"). Variations are taken with $\delta v \in L^2(\rho)$ compactly supported in time away from observation times, and $\delta\rho \in C_c^\infty((t_{k-1}, t_k)\times\mathbb{R}^d)$ with zero total mass for each t.

Lemma 6 (Variation in v (OU and elliptic))**.** *For the OU Lagrangian* $\mathcal{L} = \frac{1}{2\sigma^2}\int \|v-b\|^2\rho + \int \psi(\partial_t\rho + \nabla\cdot(\rho v) - \frac{\sigma^2}{2}\Delta\rho)$*, using Lemma 4 the first variation gives* $v - b = \sigma^2\nabla\psi$*. For the elliptic Lagrangian* $\mathcal{L} = \frac{1}{4}\int\langle v-b, D^{-1}(v-b)\rangle\rho + \int\psi(\partial_t\rho + \nabla\cdot(\rho v) - \nabla\cdot(D\nabla\rho))$*, one obtains* $v - b = 2D\nabla\psi$*.*

Lemma 7 (Variation in ρ (OU and elliptic))**.** *For variations* $\delta\rho$ *supported away from observation times and endpoints, integration by parts in time and space yields for OU*

$$\delta_\rho\mathcal{L} = \int_0^T\!\!\int \Big(\tfrac{1}{2\sigma^2}\|v-b\|^2 - \partial_t\psi - v\cdot\nabla\psi - \tfrac{\sigma^2}{2}\,\Delta\psi\Big)\delta\rho\, dx\, dt,$$

and for the elliptic case (using Lemma 4)

$$\delta_\rho\mathcal{L} = \int_0^T\!\!\int \Big(\tfrac{1}{4}\langle v-b, D^{-1}(v-b)\rangle - \partial_t\psi - v\cdot\nabla\psi - \nabla\cdot(D\nabla\psi)\Big)\delta\rho\, dx\, dt.$$

Substituting the stationarity conditions $v - b = \sigma^2\nabla\psi$ *and* $v - b = 2D\nabla\psi$ *gives the HJB equations*

$$-\partial_t\psi = \tfrac{\sigma^2}{2}\|\nabla\psi\|^2 + \langle\nabla\psi, b\rangle + \tfrac{\sigma^2}{2}\,\Delta\psi \quad (\textit{OU}), \qquad -\partial_t\psi = \tfrac{1}{2}\|\nabla\psi\|_{2D}^2 + \langle\nabla\psi, b\rangle + \nabla\cdot(D\nabla\psi) \quad (\textit{elliptic}). \tag{33}$$

Boundary and jump conditions. With hard marginal constraints at $t \in \{0, t_1, \ldots, t_K, T\}$ the admissible $\delta\rho$ vanish on these time slices, so no boundary terms arise from time integration. If, alternatively, observations are imposed softly via potentials $V_k(x) = -\log \ell(y_k \mid x)$ added to the objective as $\sum_k \int V_k(x)\, \rho_{t_k}(x)\, dx$, then allowing free $\delta\rho(t_k, \cdot)$ produces the jump conditions

$$\psi(t_k^+, x) - \psi(t_k^-, x) = V_k(x), \qquad k = 1, \ldots, K, \tag{34}$$

as claimed.

Equation (33) is precisely the HJB PDE in the main text. Eliminating $u_t = v_t - b$ between (12) and the FP constraint recovers the controlled Fokker–Planck dynamics. Hence any KL minimiser yields (ρ_t, ψ_t) solving the coupled HJB–FP system, i.e. the geodesic equation for g^{OF}.

Length–energy equivalence and geodesic characterisation. For any sufficiently regular curve $t \mapsto \rho_t$ in $\mathcal{P}_2$ and any choice of drift/diffusion (b, D), define the *control-induced tangent*

$$\mathcal{D}_t\rho_t := \partial_t\rho_t + \nabla\cdot(\rho_t b) - \nabla\cdot(D\nabla\rho_t).$$

For the OU reference, $D = \frac{\sigma^2}{2} I$ and this becomes $\mathcal{D}_t\rho_t = \partial_t\rho_t + \nabla\cdot(\rho_t b) - \frac{\sigma^2}{2}\Delta\rho_t$. Define the g^{OF}–length and energy

$$\mathrm{Len}(\rho)^2 := \Big(\int_0^T \|\mathcal{D}_t\rho_t\|_{g^{\mathrm{OF}}(\rho_t)}\, dt\Big)^2, \qquad \mathcal{E}(\rho) := \frac{1}{2}\int_0^T \|\mathcal{D}_t\rho_t\|^2_{g^{\mathrm{OF}}(\rho_t)}\, dt.$$

By Cauchy–Schwarz, minimisers of $\mathcal{E}$ under fixed endpoints coincide with constant–speed minimisers of Len. Moreover, for an admissible triple (ρ, v, u) with $u := v - b$ and the controlled FP constraint

$$\partial_t\rho + \nabla\cdot(\rho v) = \tfrac{\sigma^2}{2}\Delta\rho,$$

one has $\mathcal{D}_t\rho_t = -\nabla\cdot(\rho_t u_t)$, hence by the definition of g^{OF},

$$\|\mathcal{D}_t\rho_t\|^2_{g^{\mathrm{OF}}(\rho_t)} = \sigma^{-2} \inf_{w:\, \mathcal{D}_t\rho_t = -\nabla\cdot(\rho_t w)} \int \|w(x)\|^2\, \rho_t(x)\, dx \;\le\; \sigma^{-2}\int \|u_t(x)\|^2\, \rho_t(x)\, dx,$$

with equality when $u_t = \nabla\phi_t$ is the g^{OF}–Riesz representative of $\mathcal{D}_t\rho_t$ (in particular, for the OU–MMSB minimiser we have $u_t = \sigma^2\nabla\psi_t$ by Lemma 6). Therefore, minimising KL over admissible path measures is equivalent to minimising $\mathcal{E}(\rho)$, and the induced marginal curve is a g^{OF}–geodesic.

Corollary 5 (Geodesic characterisation under g^{OF})**.** *Let $\{\rho_t\}_{t\in[0,T]}$ be admissible with prescribed marginals at $\{0, t_1, \ldots, t_K, T\}$. Then the following are equivalent:*

1. *$\{\rho_t\}$ minimises $D_{\mathrm{KL}}(Q\|P_{\mathrm{ref}})$ among all admissible Q with these marginals (OU–MMSB solution).*
2. *There exists ψ such that (ρ, v, ψ) satisfies (12)–(33) and the weak FP constraint (30) (with jumps (34) when observations are soft).*
3. *$\{\rho_t\}$ is a constant–speed geodesic for the metric g^{OF} connecting the constrained time slices.*

Proof. (1)⇒(2) is Lemmas 6–7. (2)⇒(3) follows from $\mathcal{A} = \mathcal{E}$ and standard length–energy equivalence on Riemannian manifolds of measures (Benamou–Brenier type arguments, adapted to g^{OF}). (3)⇒(1) uses the energy minimality of constant–speed geodesics and $\mathcal{E} = \mathcal{A}$. □

Uniqueness. By Appendix J, the OU–MMSB minimiser of $D_{\mathrm{KL}}(\cdot\|P_{\mathrm{ref}})$ is unique. Via the identity (27) (equivalently, (29)), the control action $\mathcal{A}$ is strictly convex in u, hence the induced density curve is the unique geodesic compatible with the prescribed marginals (Corollary 3).

Proof of Corollary 3. By the identity (27)–(29), the unique KL minimiser coincides with the unique minimiser of $\mathcal{A}$. By the Euler–Lagrange conditions (12)–(33), this minimiser solves the HJB–FP geodesic system; hence (ρ_t) is precisely the Riemannian geodesic for (25). □

Well–posedness and regularity summary. We collect the hypotheses ensuring each step above is rigorous.

- **Existence/uniqueness of the MMSB minimiser.** Under the dominated-anchor condition $\nu_k \ll r(t_k, \cdot)$ (Appendix J) and non–degenerate OU reference, the feasible set is nonempty, tight and narrowly closed; lower semicontinuity and strict convexity of KL yield a unique minimiser.
- **Controlled Fokker–Planck well–posedness.** For globally Lipschitz drift b (in particular OU) and square-integrable controls, the controlled FP equation is well posed in the weak sense with $\rho \in C([0,T];\mathcal{P}_2)$. In the OU case, Gaussian reference tails together with finite KL provide the integrability needed for the integration by parts in Lemma 4. Standard references include dynamic OT/Benamou–Brenier and parabolic PDE texts (e.g., (Benamou & Brenier, 2000; Ambrosio et al., 2008; Villani, 2003; 2009)).
- **Girsanov–KL identity.** Lemma 5 gives $D_{\mathrm{KL}}(Q\|P_{\mathrm{ref}}) = \frac{1}{2\sigma^2}\mathbb{E}_Q \int \|u_t\|^2 dt$ under Novikov/Kazamaki ((Novikov, 1973; Kazamaki, 1994)), ensuring the Eulerian action equals the pathwise KL.
- **HJB weak solution and optimality system.** With globally Lipschitz drift b (in particular OU) and $\sigma > 0$, the semilinear parabolic HJB (33) is well–posed in the weak/viscosity sense (see (Crandall et al., 1992; Bardi & Capuzzo-Dolcetta, 1997; Fleming & Soner, 2006)); ψ is unique up to an additive constant on each open interval (t_{k-1}, t_k). The relation $v - b = \sigma^2 \nabla\psi$ holds in $L^2(\rho)$ and defines the metric velocity.
- **Observation modelling.** Hard marginal constraints enforce $\delta\rho = 0$ at observation times; soft constraints via V_k yield jumps (34). Both regimes are compatible with the weak formulation and the geodesic characterisation.

L. Proofs: Diffusion-scale limits (supplementary propositions)

Proposition 9 (Small-noise limit of the soft-potential posterior)**.** *Assume b is globally Lipschitz and each V_k is continuous and bounded below (so each likelihood weight e^{-V_k} is bounded and continuous). Assume also that the deterministic normaliser $\bar{Z}$ in (19) satisfies $0 < \bar{Z} < \infty$. Then $P^\sigma_{\mathrm{post}} \Rightarrow P^0_{\mathrm{post}}$ narrowly in $\mathcal{P}(\Omega)$ as $\sigma \to 0$. In particular, for every $t \in [0,T]$ one has $\mu^\sigma_t \Rightarrow \mu^0_t := (\Phi_t)_\# \mu^0_0$ and the posterior concentrates on deterministic prior trajectories.*

Remark. The bounded-below assumption includes common observation models where V_k has quadratic growth (e.g. Gaussian negative log-likelihoods), since only boundedness/continuity of the weights e^{-V_k} are needed for the dominated-convergence argument.

Proof sketch. By standard SDE stability, $P^\sigma_{\mathrm{ref}} \Rightarrow P^0_{\mathrm{ref}}$ where P^0_{ref} is the law of $X_t = \Phi_t(X_0)$ with $X_0 \sim \rho_0$. For any bounded continuous test functional F on Ω,

$$E_{P^\sigma_{\mathrm{post}}}[F] = \frac{E_{P^\sigma_{\mathrm{ref}}}\big[F\, e^{-\sum_k V_k(X_{t_k})}\big]}{E_{P^\sigma_{\mathrm{ref}}}\big[e^{-\sum_k V_k(X_{t_k})}\big]},$$

and both numerator and denominator converge by weak convergence of P^σ_{ref} since the weights e^{-V_k} are bounded and continuous. □

Proposition 10 (High-diffusion decoupling of the soft-potential posterior (mixing regimes))**.** *Assume each V_k is bounded, and either (i) a discrete N-point state grid with strictly positive transition matrices K^σ_k, or (ii) a compact domain with uniformly positive transition densities, such that the reference kernels become asymptotically mixing as $\sigma \to \infty$ (Doeblin/Dobrushin contraction uniformly on each segment). This excludes non-compact $\mathbb{R}^d$; we make no claim there. Then for each k,*

$$\big\|\mu^\sigma_k - \rho^{\mathrm{obs},\sigma}_{t_k}\big\|_{\mathrm{TV}} \to 0 \qquad (\sigma \to \infty),$$

and consecutive time slices become asymptotically independent in the sense that the mutual information $I_{P^\sigma_{\mathrm{post}}}(X_{t_{k-1}}; X_{t_k})$ tends to 0.

Proof sketch. Write $\mu^\sigma_k \propto \rho^{\mathrm{obs},\sigma}_{t_k}\, h^\sigma_k$ with $h^\sigma_k(x) = E_{P^\sigma_{\mathrm{ref}}}[e^{-\sum_{j\neq k} V_j(X_{t_j})} \mid X_{t_k}{=}x]$. Mixing implies the conditional law of $\{X_{t_j}\}_{j\neq k}$ given $X_{t_k}{=}x$ becomes asymptotically independent of x, hence $\mathrm{osc}(\log h^\sigma_k) \to 0$ and the TV claim follows from a bounded reweighting inequality. A discrete-grid proof (including the mutual-information claim) is given in Appendix M.

M. Proofs: High-diffusion decoupling

We give a self-contained proof for the discrete grid case (finite state space); the compact-domain case follows by replacing matrix bounds with minorization/Doeblin arguments for positive kernels.

Setup (finite grid). Let $\mathcal{X}$ be a finite state grid. For each segment $k \in \{1, \dots, K\}$ let K_k^σ be the strictly positive Markov transition matrix of the reference diffusion from t_{k-1} to t_k. Write $w_k(x) := e^{-V_k(x)}$ and assume $0 < m_k \le w_k(x) \le M_k < \infty$ on $\mathcal{X}$. The posterior on time-slices $x_{0:K}$ is

$$p_{\text{post}}^\sigma(x_{0:K}) \;\propto\; \rho_0(x_0) \prod_{k=1}^{K} K_k^\sigma(x_{k-1}, x_k)\, w_k(x_k),$$

and its marginal at t_k is denoted μ_k^σ. The reference marginal is $r_k^\sigma := (P_{\text{ref}}^\sigma)_{t_k}$, and the per-time evidence tilt is

$$\rho_{t_k}^{\text{obs},\sigma}(x) \;\propto\; r_k^\sigma(x)\, w_k(x).$$

Define the segment-wise mixing coefficients

$$\delta_k(\sigma) := \sup_{x,x' \in \mathcal{X}} \left\| K_k^\sigma(x, \cdot) - K_k^\sigma(x', \cdot) \right\|_{\text{TV}}, \qquad \delta(\sigma) := \max_{1 \le k \le K} \delta_k(\sigma).$$

The mixing assumption in Proposition 10 is $\delta(\sigma) \to 0$ as $\sigma \to \infty$.

Step 1: a bounded reweighting inequality.

Lemma 8 (TV bound under bounded log-oscillation)**.** *Let q be a probability mass function on $\mathcal{X}$ and let $h : \mathcal{X} \to (0, \infty)$. Define $p(x) := q(x)h(x)/\sum_z q(z)h(z)$ and $R := \exp(\operatorname{osc}(\log h)) = \frac{\max_x h(x)}{\min_x h(x)}$. Then*

$$\|p - q\|_{\text{TV}} \le \frac{R-1}{R+1} = \tanh\Big(\tfrac{1}{2}\, \operatorname{osc}(\log h)\Big).$$

Proof. Let $g := h/\mathbb{E}_q[h]$ so that $p = qg$ and $\mathbb{E}_q[g] = 1$. Since $\mathbb{E}_q[h] \in [\min h, \max h]$, one has $g(x) \in [1/R, R]$ for all x. The supremum of $\frac{1}{2}\mathbb{E}_q[|g - 1|] = \|p - q\|_{\text{TV}}$ over such g with mean 1 is attained by a two-point $g \in \{1/R, R\}$, giving $\|p - q\|_{\text{TV}} \le (R-1)/(R+1)$. □

Step 2: the cross-time factor becomes constant. Define the cross-time factor

$$h_k^\sigma(x) := E_{P_{\text{ref}}^\sigma}\Big[\prod_{j \ne k} w_j(X_{t_j}) \,\Big|\, X_{t_k} = x \Big].$$

Since each w_j is bounded and strictly positive, h_k^σ is bounded uniformly:

$$\prod_{j \ne k} m_j \;\le\; h_k^\sigma(x) \;\le\; \prod_{j \ne k} M_j, \qquad \forall x \in \mathcal{X}. \tag{35}$$

We claim $\operatorname{osc}(\log h_k^\sigma) \to 0$ as $\sigma \to \infty$. To this end, fix $x, x' \in \mathcal{X}$ and write P_x^σ for the conditional law of the full time-slice vector $(X_{t_0}, \dots, X_{t_K})$ under P_{ref}^σ given $X_{t_k} = x$ (and similarly $\mathsf{P}_{x'}^\sigma$). Then for $F(x_{0:K}) := \prod_{j \ne k} w_j(x_j)$ we have $0 < F \le \prod_{j \ne k} M_j$ and

$$\left| h_k^\sigma(x) - h_k^\sigma(x') \right| = \left| \mathbb{E}_{\mathsf{P}_x^\sigma}[F] - \mathbb{E}_{\mathsf{P}_{x'}^\sigma}[F] \right| \le 2\Big(\prod_{j \ne k} M_j \Big) \left\| \mathsf{P}_x^\sigma - \mathsf{P}_{x'}^\sigma \right\|_{\text{TV}}.$$

Because the reference is Markov, conditional on $X_{t_k} = x$ the past and future slices are independent and can be generated by (i) forward kernels $\{K_{k+1}^\sigma, \dots, K_K^\sigma\}$ and (ii) the time-reversal kernels $\{\widetilde{K}_k^\sigma, \dots, \widetilde{K}_1^\sigma\}$, where $\widetilde{K}_j^\sigma(x_j, x_{j-1}) := r_{j-1}^\sigma(x_{j-1}) K_j^\sigma(x_{j-1}, x_j)/r_j^\sigma(x_j)$. Let $\widetilde{\delta}_j(\sigma)$ be the Dobrushin coefficient of $\widetilde{K}_j^\sigma$. A finite-horizon coupling bound yields, uniformly in x, x',

$$\left\| \mathsf{P}_x^\sigma - \mathsf{P}_{x'}^\sigma \right\|_{\text{TV}} \le \sum_{j=1}^{k} \widetilde{\delta}_j(\sigma) + \sum_{j=k+1}^{K} \delta_j(\sigma).$$

Under strict positivity and $\delta_j(\sigma) \to 0$, one also has $\widetilde{\delta}_j(\sigma) \to 0$ (rows of K_j^σ become source-independent, hence the reversal becomes destination-independent). Consequently $\sup_{x,x'} |h_k^\sigma(x) - h_k^\sigma(x')| \to 0$; combining with the uniform lower bound in (35) yields $\operatorname{osc}(\log h_k^\sigma) \to 0$.

Step 3: marginal decoupling. By Proposition 3 applied to P_{ref}^σ,

$$\mu_k^\sigma(x) \;\propto\; r_k^\sigma(x)\, w_k(x)\, h_k^\sigma(x) \;\propto\; \rho_{t_k}^{\mathrm{obs},\sigma}(x)\, h_k^\sigma(x).$$

Lemma 8 with $q = \rho_{t_k}^{\mathrm{obs},\sigma}$ and $h = h_k^\sigma$ gives $\|\mu_k^\sigma - \rho_{t_k}^{\mathrm{obs},\sigma}\|_{\mathrm{TV}} \le \tanh(\frac{1}{2}\operatorname{osc}(\log h_k^\sigma)) \to 0$. This proves the TV decoupling claim in Proposition 10.

Step 4: mutual information. Let $\pi_k^\sigma = (P_{\mathrm{post}}^\sigma)_{t_{k-1},t_k}$. Define the backward message $\beta_k^\sigma(x) := E_{P_{\mathrm{ref}}^\sigma}[\prod_{j>k} w_j(X_{t_j}) \mid X_{t_k}{=}x]$, so that $m_{>k} \le \beta_k^\sigma \le M_{>k}$ for $m_{>k} := \prod_{j>k} m_j$ and $M_{>k} := \prod_{j>k} M_j$. By Markov structure, the posterior one-step conditional is a bounded reweighting of the reference kernel:

$$P_{\mathrm{post}}^\sigma(X_{t_k}{=}y \mid X_{t_{k-1}}{=}x) = \frac{K_k^\sigma(x,y)\, w_k(y)\, \beta_k^\sigma(y)}{\sum_{z\in\mathcal{X}} K_k^\sigma(x,z)\, w_k(z)\, \beta_k^\sigma(z)}.$$

Since $w_k\beta_k^\sigma$ has bounded ratio on $\mathcal{X}$, this reweighting does not amplify Dobrushin coefficients by more than a constant factor; hence $\sup_{x,x'} \|P_{\mathrm{post}}^\sigma(X_{t_k} \in \cdot \mid X_{t_{k-1}}{=}x) - P_{\mathrm{post}}^\sigma(X_{t_k} \in \cdot \mid X_{t_{k-1}}{=}x')\|_{\mathrm{TV}} \to 0$ as $\sigma \to \infty$. Therefore each row of the posterior conditional kernel converges in TV to the common mixture μ_k^σ, and

$$\left\|\pi_k^\sigma - \mu_{k-1}^\sigma \otimes \mu_k^\sigma\right\|_{\mathrm{TV}} = \mathbb{E}_{\mu_{k-1}^\sigma}\Big[\left\|P_{\mathrm{post}}^\sigma(X_{t_k} \in \cdot \mid X_{t_{k-1}}) - \mu_k^\sigma\right\|_{\mathrm{TV}}\Big] \to 0.$$

Since $\mathcal{X}$ is finite and all slice probabilities are strictly positive, this TV convergence implies $D_{\mathrm{KL}}(\pi_k^\sigma \| \mu_{k-1}^\sigma \otimes \mu_k^\sigma) \to 0$ by continuity of KL on the simplex interior, i.e. $I_{P_{\mathrm{post}}^\sigma}(X_{t_{k-1}}; X_{t_k}) \to 0$.

N. Algorithms: Posterior Doob smoother and IPFP

This appendix describes two complementary routines on discretised grids: (i) exact posterior computation via Doob–h / forward–backward message passing (fixed likelihood potentials), and (ii) multi-marginal IPFP for *exogenous*-anchor Schrödinger bridges used in controlled geometry diagnostics.

N.1. Exact posterior on a grid (Doob–h / forward–backward)

N.1.1. Posterior factorisation

For discrete observations $y_{1:K}$, the posterior on time-slices admits the Markov factorisation

$$p_{\mathrm{post}}(x_{0:K}) \;\propto\; r(t_0, x_0) \prod_{k=1}^{K} K_{t_{k-1}\to t_k}(x_{k-1}, x_k) \prod_{k=1}^{K} \ell(y_k \mid x_k). \tag{36}$$

Equivalently, P_{post} is a Doob–h transform of the reference diffusion with time-sliced potentials $\ell(y_k \mid \cdot)$.

N.1.2. Log-domain forward–backward

On a grid, the posterior marginals μ_k are computed by the standard forward–backward recursion. Define the (unnormalised) forward message

$$\vec{\alpha}_k(x_k) := \ell(y_k \mid x_k) \int K_{t_{k-1}\to t_k}(x_{k-1}, x_k)\, \vec{\alpha}_{k-1}(x_{k-1})\, \mathrm{d}x_{k-1}, \qquad \vec{\alpha}_0(x_0) := r(t_0, x_0),$$

and the backward message

$$\overleftarrow{\beta}_{k-1}(x_{k-1}) := \int K_{t_{k-1}\to t_k}(x_{k-1}, x_k)\, \ell(y_k \mid x_k)\, \overleftarrow{\beta}_k(x_k)\, \mathrm{d}x_k, \qquad \overleftarrow{\beta}_K(x_K) := 1.$$

Then $\mu_k(x) \propto \vec{\alpha}_k(x)\, \overleftarrow{\beta}_k(x)$. We implement these recursions in log-space with log-sum-exp normalisation.

N.2. Exogenous-anchor bridges via IPFP (diagnostic)

N.2.1. Schrödinger system and IPFP updates

When one prescribes exogenous target marginals $\{\rho_{t_k}^{\text{tar}}\}_{k=0}^{K}$ and solves the bridge

$$Q^\star = \arg \min_{Q:\, Q_{t_k}=\rho_{t_k}^{\text{tar}}\ \forall k} D_{\text{KL}}(Q\|P_{\text{ref}}),$$

the unique solution admits the Schrödinger system form

$$q^\star(x_0,\dots,x_K) = \phi_0(x_0)\, r(t_0,x_0) \prod_{k=1}^{K} K_{t_{k-1}\to t_k}(x_{k-1},x_k)\, \phi_k(x_k), \tag{37}$$

where $\{\phi_k\}_{k=0}^{K}$ are positive potentials chosen so that the marginals of $q^\star$ match $\rho_{t_k}^{\text{tar}}$. Multi-marginal IPFP is a cyclic KL projection method to find these potentials.

Forward-Backward Messages. Let $\vec{\alpha}_k(x_k)$ be the forward message at time t_k and $\overleftarrow{\beta}_k(x_k)$ be the backward message at time t_k. They are computed recursively in the log-domain:

$$\log \vec{\alpha}_k(x_k) = \log \int K_{t_{k-1}\to t_k}(x_{k-1},x_k)\ \exp\!\big(\log \vec{\alpha}_{k-1}(x_{k-1}) + \log \phi_{k-1}^{(j)}(x_{k-1})\big)\, \mathrm{d}x_{k-1}, \tag{38}$$

$$\log \overleftarrow{\beta}_k(x_k) = \log \int K_{t_k\to t_{k+1}}(x_k,x_{k+1})\ \exp\!\big(\log \overleftarrow{\beta}_{k+1}(x_{k+1}) + \log \phi_{k+1}^{(j)}(x_{k+1})\big)\, \mathrm{d}x_{k+1}, \tag{39}$$

with initial conditions $\log \vec{\alpha}_0(x_0) = \log r(t_0,x_0)$ and $\log \overleftarrow{\beta}_K(x_K) = 0$.

Marginal Computation and Update. The current marginal at time t_k, given the potentials, is proportional to the product of the forward message, backward message, and the potential itself. In the log-domain:

$$\log \tilde{\rho}_{t_k}(x_k) = \log \vec{\alpha}_k(x_k) + \log \overleftarrow{\beta}_k(x_k) + \log \phi_k^{(j)}(x_k) + C_k, \tag{40}$$

where C_k is a normalization constant. The IPFP update for $\log \phi_k$ that enforces the marginal constraint is then:

$$\log \phi_k^{(j+1)}(x_k) \leftarrow \log \phi_k^{(j)}(x_k) + \big(\log \rho_{t_k}^{\text{tar}}(x_k) - \log \tilde{\rho}_{t_k}(x_k)\big)\,. \tag{41}$$

This simplifies to $\log \phi_k^{(j+1)}(x_k) \leftarrow \log \rho_{t_k}^{\text{tar}}(x_k) - \log \vec{\alpha}_k(x_k) - \log \overleftarrow{\beta}_k(x_k)$. The full algorithm is detailed in Algorithm 1.

Implementation note (log normalisation). After each forward/backward sweep we compute the constant $C_k^{(j)} = \log \sum_x \exp\!\big(\log \vec{\alpha}_k(x) + \log \overleftarrow{\beta}_k(x) + \log \phi_k^{(j)}(x)\big)$ for every time index k and subtract it from $\log \tilde{\rho}_{t_k}$. This log-sum-exp centring keeps messages numerically stable and prevents overflow when working with high-dimensional grids.

O. Algorithms: IPFP convergence (proof sketch; strictly positive kernels)

Lemma 9 (Multi-marginal IPFP convergence (discrete, strictly positive kernels))**.** *On a finite state grid with strictly positive reference kernels (Doeblin condition), the multi-marginal IPFP algorithm satisfies:*

1. ***Monotonic Descent:*** *The KL divergence to the optimal solution Q^* satisfies $D_{\text{KL}}(Q^*\|Q^{(j+1)}) \le D_{\text{KL}}(Q^*\|Q^{(j)})$ at each projection step.*

2. ***Linear Convergence:*** *Under (48), the potentials contract linearly in the Hilbert projective metric; on a finite grid this implies linear convergence of the iterates (e.g. $D_{\text{KL}}(Q^*\|Q^{(j)}) \to 0$ at a linear rate).*

This appendix sketches the argument and records the key ingredients (Csiszár I-projections and Birkhoff/Hilbert-metric contraction). For full generality and sharp rates under entropic regularisation, see, e.g., Csiszár (1975); Franklin & Lorenz (1989); Lemmens & Nussbaum (2012); Carlier et al. (2023).

Algorithm 1 Detailed Log-Domain Multi-Marginal IPFP

Input: Target log-marginals $\{\log \rho_{t_k}^{\text{tar}}\}_{k=0}^{K}$, reference log-marginal $\log r(t_0, \cdot)$, transition kernels $\{K_{t_{k-1}\to t_k}\}_{k=1}^{K}$, tolerance $\delta > 0$, max iterations M.
Initialize: Log-potentials $\log \phi_k^{(0)}(x) \leftarrow 0$ for all x, k. Iteration $j \leftarrow 0$.
repeat
  $\log \phi^{\text{prev}} \leftarrow \log \phi^{(j)}$.
  for $k = 0$ **to** K **do**
    // Forward messages only if $k \geq 1$ (identity for $k = 0$)
    $\log \vec{\alpha}_0(x_0) \leftarrow \log r(t_0, x_0)$.
    for $i = 1$ **to** k **do**
      $\log \vec{\alpha}_i(x_i) \leftarrow \log \int K_{t_{i-1}\to t_i}(x_{i-1}, x_i)\ \exp\big(\log \vec{\alpha}_{i-1}(x_{i-1}) + \log \phi_{i-1}^{(j)}(x_{i-1})\big)\, \mathrm{d}x_{i-1}$.
    end for
    // Compute all backward messages up to t_k
    $\log \overleftarrow{\beta}_K(x_K) \leftarrow 0$.
    for $i = K - 1$ **down to** k **do**
      $\log \overleftarrow{\beta}_i(x_i) \leftarrow \log \int K_{t_i\to t_{i+1}}(x_i, x_{i+1})\ \exp\big(\log \overleftarrow{\beta}_{i+1}(x_{i+1}) + \log \phi_{i+1}^{(j)}(x_{i+1})\big)\, \mathrm{d}x_{i+1}$.
    end for
    // Update log-potential $\log \phi_k$
    $\log \phi_k^{(j+1)}(x) \leftarrow \log \rho_{t_k}^{\text{tar}}(x) - \log \vec{\alpha}_k(x) - \log \overleftarrow{\beta}_k(x)$.
    To ensure stability, other potentials $\log \phi_{i\neq k}^{(j+1)} \leftarrow \log \phi_{i\neq k}^{(j)}$.
  end for
  $j \leftarrow j + 1$.
  // Convergence check on the change in potentials
  $\epsilon^{(j)} \leftarrow \max_k \| \log \phi_k^{(j)} - \log \phi_k^{\text{prev}} \|_{L^\infty}$.
until $\epsilon^{(j)} < \delta$ **or** $j \geq M$
Return: Converged log-potentials $\{\log \phi_k^{(j)}\}_{k=0}^{K}$.

O.0.1. Preliminaries: I-Projections and the Pythagorean Identity

The foundation of the monotonic descent proof lies in the geometric properties of the Kullback-Leibler (KL) divergence. The KL divergence, also known as relative entropy, is a member of a class of divergences known as Bregman divergences. This geometric structure is key.

Definition 3 (I-Projection)**.** Let $\mathcal{P}(\mathcal{X})$ be the space of probability measures on a set $\mathcal{X}$. Let $\mathcal{C} \subset \mathcal{P}(\mathcal{X})$ be a closed, convex set of probability measures. The **I-projection** (Information Projection) of a measure $Q \in \mathcal{P}(\mathcal{X})$ onto $\mathcal{C}$ is the unique measure $P^* \in \mathcal{C}$ that minimizes the KL divergence from Q:

$$P^* = I_{\mathcal{C}}(Q) := \arg\min_{P \in \mathcal{C}} D_{\mathrm{KL}}(P \| Q). \tag{42}$$

The existence and uniqueness of the I-projection are guaranteed if $\mathcal{C}$ is closed and convex. The central property of I-projections is the following "Pythagorean" identity, due to Csiszár.

Lemma 10 (Generalized Pythagorean Identity (Csiszár, 1975))**.** *Let $\mathcal{C}$ be a closed, convex set of probability measures. For any measure $P \in \mathcal{C}$ and any measure Q for which the I-projection $I_{\mathcal{C}}(Q)$ exists, the following identity holds:*

$$D_{\mathrm{KL}}(P \| Q) = D_{\mathrm{KL}}(P \| I_{\mathcal{C}}(Q)) + D_{\mathrm{KL}}(I_{\mathcal{C}}(Q) \| Q). \tag{43}$$

The identity holds with equality because the constraint sets $\mathcal{C}_k$ are affine subspaces of the space of measures, and our reference measure P_{ref} (and subsequent iterates) have sufficient support relative to the target marginals.

O.0.2. Proof of Monotonic Descent

We now apply this machinery to the Multi-Marginal IPFP algorithm. Each step of the algorithm can be seen as an I-projection onto a constraint set.

Define the Constraint Sets. For each $k \in \{0, 1, \ldots, K\}$, let $\mathcal{C}_k$ be the set of path measures that satisfy the k-th marginal constraint:

$$\mathcal{C}_k := \{P \in \mathcal{P}(\Omega) \mid P_{t_k} = \rho^{\mathrm{tar}}_{t_k}\}. \tag{44}$$

Each $\mathcal{C}_k$ is an affine, and therefore closed and convex, subset of $\mathcal{P}(\Omega)$. The unique solution to the MMSB problem, Q^*, lies in the intersection of all these sets: $Q^* \in \bigcap_{k=0}^{K} \mathcal{C}_k$.

Frame IPFP as a Sequence of I-Projections. The IPFP algorithm is a cyclic procedure. Let $Q^{(j)}$ be the path measure at the beginning of a cycle. The algorithm sequentially updates the measure to satisfy each constraint. Let's consider the update for constraint k. The updated measure, which we denote $Q^{(j+1)}$, is constructed to satisfy $Q^{(j+1)}_{t_k} = \rho^{\mathrm{tar}}_{t_k}$ while being "closest" to $Q^{(j)}$ in the KL sense. This is precisely an I-projection. Specifically, the update step for the k-th potential corresponds to finding a new path measure $Q^{(j+1)}$ such that:

$$Q^{(j+1)} = \arg\min_{P \in \mathcal{C}_k} D_{\mathrm{KL}}(P \| Q^{(j)}) = I_{\mathcal{C}_k}(Q^{(j)}). \tag{45}$$

Apply the Pythagorean Identity. Let us apply Lemma 10. We choose our three points as:

- The true solution $Q^* \in \mathcal{C}_k$.
- The current iterate $Q^{(j)}$.
- The next iterate $Q^{(j+1)} = I_{\mathcal{C}_k}(Q^{(j)})$.

Substituting these into the Pythagorean identity gives:

$$D_{\mathrm{KL}}(Q^* \| Q^{(j)}) = D_{\mathrm{KL}}(Q^* \| Q^{(j+1)}) + D_{\mathrm{KL}}(Q^{(j+1)} \| Q^{(j)}). \tag{46}$$

Since $D_{\mathrm{KL}}(Q^{(j+1)} \| Q^{(j)}) \geq 0$, we immediately have the desired result:

$$D_{\mathrm{KL}}(Q^* \| Q^{(j)}) \geq D_{\mathrm{KL}}(Q^* \| Q^{(j+1)}). \tag{47}$$

This shows that the KL divergence to the true solution is monotonically non-increasing at every single step of the IPFP cycle. Strict inequality holds unless $Q^{(j+1)} = Q^{(j)}$, which occurs if and only if $Q^{(j)}$ already satisfies the constraint k (i.e., $Q^{(j)} \in \mathcal{C}_k$). Over a full cycle of $K+1$ projections, the KL divergence must strictly decrease unless the algorithm has converged to Q^*, which is the only state satisfying all constraints simultaneously. This completes the proof of monotonic descent.

O.0.3. Proof of Linear Convergence Rate

The proof of linear convergence relies on analyzing the IPFP operator's action on the potentials in the Hilbert projective metric, leveraging the theory of contraction mappings on cones.

Doeblin Condition and Uniform Positivity. Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable state space. We assume that the transition kernels $K_{t_{k-1}\to t_k}(x, y)$ satisfy a *uniform positivity* (Doeblin) condition: there exist constants $0 < m_k \leq M_k < \infty$ such that

$$m_k \leq K_{t_{k-1}\to t_k}(x, y) \leq M_k, \quad \forall (x, y) \in \mathcal{X} \times \mathcal{X}. \tag{48}$$

The ratio $\eta_k := m_k/M_k \in (0, 1]$ serves as a lower bound on the mixing of the kernel. This condition is satisfied for any discretized kernel with non-zero entries, or for Gaussian kernels on a compact domain.

The Hilbert Projective Metric. Let $\mathcal{C}_+(\mathcal{X})$ be the cone of strictly positive, measurable functions on $\mathcal{X}$. For any two functions $f, g \in \mathcal{C}_+(\mathcal{X})$, the Hilbert projective metric is defined as:

$$d_H(f, g) := \log \left(\frac{\sup_{x\in\mathcal{X}}(f(x)/g(x))}{\inf_{x\in\mathcal{X}}(f(x)/g(x))} \right). \tag{49}$$

This is a metric on the projective space of rays in $\mathcal{C}_+$, meaning $d_H(f, g) = 0$ if and only if $f = cg$ for some $c > 0$.

The IPFP Operator as a Contraction. Each IPFP update for a potential ϕ_k can be viewed as an operator T_k acting on the other potentials. Specifically, updating ϕ_k to satisfy the k-th marginal constraint ρ_k^{tar} involve integral operators $\mathcal{K}$ with kernels K. A fundamental result by Birkhoff (1957) states that such integral operators are strict contractions in the Hilbert metric.

Lemma 11 (Birkhoff's Contraction Theorem)**.** *Let $\mathcal{K}$ be an integral operator with a kernel satisfying the Doeblin condition (48). Then $\mathcal{K}$ is a strict contraction on $\mathcal{C}_+$ with respect to d_H, with contraction coefficient*

$$\tau_k = \frac{1 - \sqrt{\eta_k}}{1 + \sqrt{\eta_k}} = \tanh\left(\frac{1}{4}\log\frac{1}{\eta_k^2}\right) < 1. \tag{50}$$

A full cycle of the Multi-Marginal IPFP algorithm corresponds to a composition of such operators. Since the inversion map $f \mapsto 1/f$ and multiplication by fixed positive marginals $f \mapsto \rho \cdot f$ are isometries in the Hilbert metric, the composite operator $T = T_K \circ \cdots \circ T_0$ is a strict contraction with coefficient $\alpha \leq \prod_{k=1}^K \tau_k < 1$. By the Banach Fixed Point Theorem, the vector of potentials $\Psi = (\phi_0, \ldots, \phi_K)$ converges linearly to the unique fixed point Ψ^*:

$$d_H(\Psi^{(j+1)}, \Psi^*) \leq \alpha \cdot d_H(\Psi^{(j)}, \Psi^*). \tag{51}$$

Quantitative Conversion to KL Divergence. We now provide a quantitative bound connecting the Hilbert metric convergence of potentials to the KL divergence of the path measures. Let $q^{(j)}$ be the density of the iterate at cycle j, and q^* be the optimal density. From the Schrödinger system form (37), we have:

$$\log\frac{q^{(j)}(x)}{q^*(x)} = \sum_{k=0}^{K}\left(\log\phi_k^{(j)}(x_k) - \log\phi_k^*(x_k)\right) + C^{(j)}, \tag{52}$$

where $C^{(j)}$ is a normalization constant. Let $\epsilon_k^{(j)} := d_H(\phi_k^{(j)}, \phi_k^*)$. By definition of d_H, there exist constants $a_k^{(j)}$ such that $|\log\phi_k^{(j)}(x_k) - \log\phi_k^*(x_k) - a_k^{(j)}| \leq \epsilon_k^{(j)}$ for all $x_k \in \mathcal{X}$. Summing these, we obtain:

$$\left|\log\frac{q^{(j)}(x)}{q^*(x)} - A^{(j)}\right| \leq \sum_{k=0}^{K}\epsilon_k^{(j)} =: E^{(j)}, \tag{53}$$

where $A^{(j)} = C^{(j)} + \sum a_k^{(j)}$. Since $\int q^{(j)} dx = \int q^* dx = 1$, the constant $A^{(j)}$ must satisfy $|A^{(j)}| \le E^{(j)}$. Consequently, the log-density ratio is uniformly bounded:

$$\left\| \log \frac{q^{(j)}}{q^*} \right\|_\infty \le 2E^{(j)}. \tag{54}$$

Using the property that $D_{\mathrm{KL}}(Q\|P) \le \|\log(dQ/dP)\|_\infty$ for any probability measures Q, P, we conclude:

$$D_{\mathrm{KL}}(Q^* \| Q^{(j)}) \le 2E^{(j)} \le 2(K+1)C\alpha^j. \tag{55}$$

This establishes the global linear convergence of the KL divergence with the same rate α as the Hilbert metric contraction.

Positivity assumption. The Hilbert–metric contraction argument above requires every discretised kernel $\mathbf{K}_{s\to t}$ to be *strictly positive* ($\min_{i,j} K_{ij} > 0$). In practice one enforces this by an ε–regulariser, $\mathbf{K} \leftarrow (1-\varepsilon)\mathbf{K} + \varepsilon/N$, or by choosing Gaussian bandwidths that guarantee positivity; see Carlier et al. (2023) for convergence rates under such regularisation.

P. Detailed Derivation of the Connection to the RTS Smoother

Proposition 11 (MMSB SMOOTHER Recovers RTS)**.** *In the linear-Gaussian setting with an OU reference process and Gaussian observations, the exact Bayesian posterior computed by our Doob forward–backward recursion coincides exactly with the classical Rauch–Tung–Striebel smoother.*

This appendix provides a full derivation of the equivalence stated above.

P.0.1. THE LINEAR-GAUSSIAN SETTING

We consider the specific case where:

1. The reference process P_{ref} is a linear SDE (an Ornstein-Uhlenbeck process), as in (1).
2. The prior distribution ρ_0 is a Gaussian, $\mathcal{N}(\mu_0, P_0)$.
3. The observation likelihoods $\ell(y_k|x_k)$ correspond to a linear observation model with Gaussian noise:
$$y_k = C_k x_k + v_k, \quad v_k \sim \mathcal{N}(0, R_k). \tag{56}$$
This implies that the likelihood is $\ell(y_k|x_k) \propto \exp(-\frac{1}{2}\|y_k - C_k x_k\|^2_{R_k^{-1}})$.

Under these conditions, the OU reference induces Gaussian transition kernels between observation times. Writing $\Delta t_k := t_k - t_{k-1}$, one has

$$X_{t_k} \mid X_{t_{k-1}} = x \;\sim\; \mathcal{N}(F_k x,\, Q_k), \qquad F_k := e^{-A\Delta t_k}, \quad Q_k := \int_0^{\Delta t_k} e^{-As}\, \Sigma\Sigma^\top\, e^{-A^\top s}\, \mathrm{d}s.$$

Together with the Gaussian likelihood factors, the time-slice posterior (36) is a linear–Gaussian state-space model and hence jointly Gaussian.

P.0.2. FORWARD–BACKWARD EQUALS RTS

Let (m_k^-, P_k^-) denote the prediction at time t_k, (m_k^f, P_k^f) the filtered moments, and (m_k^s, P_k^s) the smoothed moments of X_{t_k}. The forward pass is the Kalman filter:

$$\begin{aligned} m_k^- &= F_k m_{k-1}^f, \qquad P_k^- = F_k P_{k-1}^f F_k^\top + Q_k, \\ S_k &= C_k P_k^- C_k^\top + R_k, \qquad G_k = P_k^- C_k^\top S_k^{-1}, \\ m_k^f &= m_k^- + G_k(y_k - C_k m_k^-), \qquad P_k^f = P_k^- - G_k S_k G_k^\top. \end{aligned}$$

The backward pass is the Rauch–Tung–Striebel smoother (Rauch et al., 1965; Särkkä, 2013):

$$\begin{aligned} J_k &= P_k^f F_{k+1}^\top (P_{k+1}^-)^{-1}, \\ m_k^s &= m_k^f + J_k(m_{k+1}^s - m_{k+1}^-), \\ P_k^s &= P_k^f + J_k(P_{k+1}^s - P_{k+1}^-) J_k^\top. \end{aligned}$$

Our posterior computation is precisely the forward–backward recursion (Appendix N); in the linear–Gaussian case, Gaussian closure yields the RTS equations above. Hence MMSB SMOOTHER recovers RTS exactly.